

(12) **United States Patent**
Kleinbeck et al.

(10) **Patent No.:** **US 12,387,608 B1**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **UNMANNED VEHICLE RECOGNITION AND
THREAT MANAGEMENT**

(71) Applicant: **Digital Global Systems, Inc.**, Tysons
Corner, VA (US)

(72) Inventors: **David William Kleinbeck**, Lees
Summit, MO (US); **Armando**
Montalvo, Winter Garden, FL (US)

(73) Assignee: **Digital Global Systems, Inc.**, Tysons
Corner, VA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **19/098,376**

(22) Filed: **Apr. 2, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/985,777, filed on
Dec. 18, 2024, now Pat. No. 12,307,905, which is a
(Continued)

(51) **Int. Cl.**
G08G 5/22 (2025.01)
G01S 3/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G08G 5/22** (2025.01); **G01S 3/043**
(2013.01); **G01S 3/046** (2013.01); **G01S 3/46**
(2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G01S 3/043; G01S 3/046; G01S 3/46;
G06N 3/08; G06N 7/01; G06N 20/00;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,215,345 A 7/1980 Robert et al.
4,400,700 A 8/1983 Rittenbach
(Continued)

FOREIGN PATENT DOCUMENTS

CN 110035440 A 7/2019
CN 115482071 A 12/2022
(Continued)

OTHER PUBLICATIONS

“A Hardware Design for Time Delay Estimation of TDOA”; Li et
al.; 2013 IEEE International Conference on Signal Processing,
Communication and Computing (ICSPCC 2013); Aug. 2013 (Year:
2013).

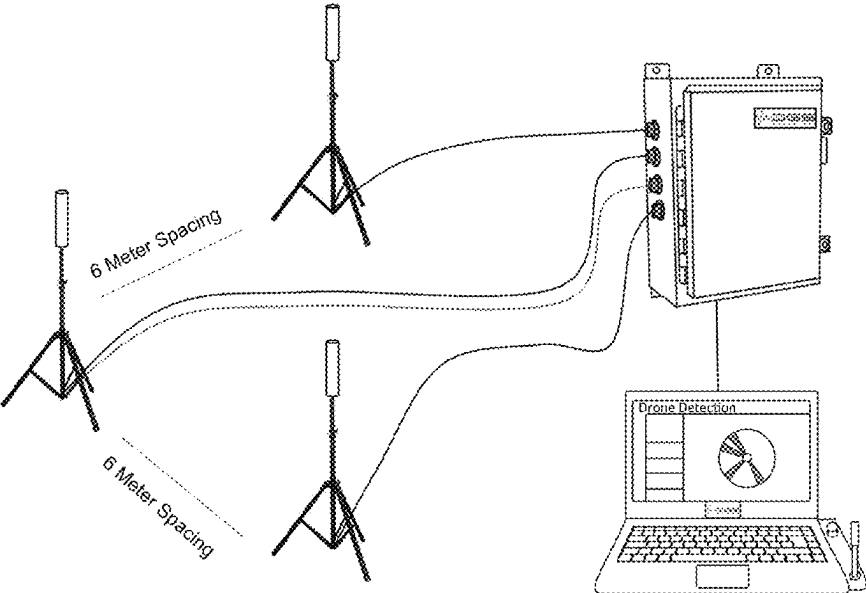
(Continued)

Primary Examiner — Nguyen T Vo
(74) *Attorney, Agent, or Firm* — NEO IP

(57) **ABSTRACT**

Systems and methods for automated unmanned aerial
vehicle recognition. A multiplicity of receivers captures RF
data and transmits the RF data to at least one node device.
The at least one node device comprises a signal processing
engine, a detection engine, a classification engine, and a
direction finding engine. The at least one node device is
configured with an artificial intelligence algorithm. The
detection engine and classification engine are trained to
detect and classify signals from unmanned vehicles and their
controllers based on processed data from the signal process-
ing engine. The direction finding engine is operable to
provide lines of bearing for detected unmanned vehicles.

20 Claims, 23 Drawing Sheets



Related U.S. Application Data

- continuation of application No. 18/789,011, filed on Jul. 30, 2024, now Pat. No. 12,243,431, which is a continuation of application No. 18/785,936, filed on Jul. 26, 2024, now Pat. No. 12,205,477, which is a continuation-in-part of application No. 18/775,710, filed on Jul. 17, 2024, now Pat. No. 12,183,213, which is a continuation-in-part of application No. 18/428,606, filed on Jan. 31, 2024, now Pat. No. 12,266,272, which is a continuation of application No. 18/374,376, filed on Sep. 28, 2023, now Pat. No. 11,893,893, which is a continuation of application No. 18/142,904, filed on May 3, 2023, now Pat. No. 11,783,712, which is a continuation of application No. 17/991,348, filed on Nov. 21, 2022, now Pat. No. 11,645,921, which is a continuation of application No. 17/735,615, filed on May 3, 2022, now Pat. No. 11,521,498, which is a continuation of application No. 17/190,048, filed on Mar. 2, 2021, now Pat. No. 11,328,609, which is a continuation of application No. 16/732,811, filed on Jan. 2, 2020, now Pat. No. 10,943,493, which is a continuation of application No. 16/275,575, filed on Feb. 14, 2019, now Pat. No. 10,529,241, which is a continuation-in-part of application No. 16/274,933, filed on Feb. 13, 2019, now Pat. No. 10,498,951, which is a continuation-in-part of application No. 16/180,690, filed on Nov. 5, 2018, now Pat. No. 10,459,020, which is a continuation-in-part of application No. 15/412,982, filed on Jan. 23, 2017, now Pat. No. 10,122,479.
- (60) Provisional application No. 62/722,420, filed on Aug. 24, 2018, provisional application No. 62/632,276, filed on Feb. 19, 2018.
- (51) **Int. Cl.**
G01S 3/46 (2006.01)
G06N 3/08 (2023.01)
G06N 7/01 (2023.01)
G08G 5/55 (2025.01)
G08G 5/57 (2025.01)
- (52) **U.S. Cl.**
CPC **G06N 3/08** (2013.01); **G06N 7/01** (2023.01); **G08G 5/55** (2025.01); **G08G 5/57** (2025.01)
- (58) **Field of Classification Search**
CPC G06N 20/10; G08G 5/22; G08G 5/55; G08G 5/57; H04W 72/04; H04W 76/11; H04W 76/15; H04W 76/23; H04W 84/12
See application file for complete search history.
- (56) **References Cited**
U.S. PATENT DOCUMENTS
- | | | | |
|-----------|----|---------|------------------------|
| 4,453,137 | A | 6/1984 | Rittenbach |
| 4,501,020 | A | 2/1985 | Wakeman |
| 4,581,769 | A | 4/1986 | Grimsley et al. |
| 4,638,493 | A | 1/1987 | Bishop et al. |
| 4,794,325 | A | 12/1988 | Britton et al. |
| 4,928,106 | A | 5/1990 | Ashjaee et al. |
| 5,103,402 | A | 4/1992 | Morton et al. |
| 5,134,407 | A | 7/1992 | Lorenz et al. |
| 5,144,642 | A | 9/1992 | Weinberg et al. |
| 5,166,664 | A | 11/1992 | Fish |
| 5,230,087 | A | 7/1993 | Meyer et al. |
| 5,293,170 | A | 3/1994 | Lorenz et al. |
| 5,343,212 | A | 8/1994 | Rose et al. |
| 5,393,713 | A | 2/1995 | Schwob |
| 5,416,593 | A | 5/1995 | Vercruysse |
| 5,448,309 | A | 9/1995 | Won |
| 5,506,864 | A | 4/1996 | Schilling |
| 5,513,385 | A | 4/1996 | Tanaka |
| 5,548,809 | A | 8/1996 | Lemson |
| 5,570,099 | A | 10/1996 | DesJardins |
| 5,589,835 | A | 12/1996 | Gildea et al. |
| 5,612,703 | A | 3/1997 | Mallinckrodt |
| 5,642,732 | A | 7/1997 | Wang |
| 5,831,874 | A | 11/1998 | Boone et al. |
| 5,835,857 | A | 11/1998 | Otten |
| 5,838,906 | A | 11/1998 | Doyle et al. |
| 5,846,208 | A | 12/1998 | Pichlmayr et al. |
| 5,856,803 | A | 1/1999 | Pevler |
| 5,936,575 | A | 8/1999 | Azzarelli et al. |
| 6,018,312 | A | 1/2000 | Haworth |
| 6,039,692 | A | 3/2000 | Kristoffersen |
| 6,085,090 | A | 7/2000 | Yee et al. |
| 6,115,580 | A | 9/2000 | Chuprun et al. |
| 6,134,445 | A | 10/2000 | Gould et al. |
| 6,144,336 | A | 11/2000 | Preston et al. |
| 6,157,619 | A | 12/2000 | Ozluturk et al. |
| 6,160,511 | A | 12/2000 | Pfeil et al. |
| 6,167,277 | A | 12/2000 | Kawamoto |
| 6,185,309 | B1 | 2/2001 | Attias |
| 6,188,715 | B1 | 2/2001 | Partyka |
| 6,191,731 | B1 | 2/2001 | McBurney et al. |
| 6,198,414 | B1 | 3/2001 | McPherson et al. |
| 6,249,252 | B1 | 6/2001 | Dupray |
| 6,286,021 | B1 | 9/2001 | Tran et al. |
| 6,296,612 | B1 | 10/2001 | Mo et al. |
| 6,304,760 | B1 | 10/2001 | Thomson et al. |
| 6,314,366 | B1 | 11/2001 | Farmakis et al. |
| 6,339,396 | B1 | 1/2002 | Mayersak |
| 6,384,776 | B1 | 5/2002 | Martin |
| 6,400,647 | B1 | 6/2002 | Huntress |
| 6,418,131 | B1 | 7/2002 | Snelling et al. |
| 6,430,418 | B1 | 8/2002 | Nivens et al. |
| 6,433,671 | B1 | 8/2002 | Nysen |
| 6,490,318 | B1 | 12/2002 | Larsson et al. |
| 6,492,945 | B2 | 12/2002 | Counselman, III et al. |
| 6,512,788 | B1 | 1/2003 | Kuhn et al. |
| 6,628,231 | B2 | 9/2003 | Mayersak |
| 6,677,895 | B1 | 1/2004 | Holt |
| 6,697,439 | B1 | 2/2004 | Trivedi et al. |
| 6,707,910 | B1 | 3/2004 | Valve et al. |
| 6,711,404 | B1 | 3/2004 | Arpee et al. |
| 6,741,595 | B2 | 5/2004 | Maher et al. |
| 6,771,957 | B2 | 8/2004 | Chitrapu |
| 6,785,321 | B1 | 8/2004 | Yang et al. |
| 6,850,557 | B1 | 2/2005 | Gronemeyer |
| 6,850,735 | B2 | 2/2005 | Sugar et al. |
| 6,859,831 | B1 | 2/2005 | Gelvin et al. |
| 6,861,982 | B2 | 3/2005 | Forstrom et al. |
| 6,876,326 | B2 | 4/2005 | Martorana |
| 6,898,197 | B1 | 5/2005 | Lavean |
| 6,898,235 | B1 | 5/2005 | Carlin et al. |
| 6,904,269 | B1 | 6/2005 | Deshpande et al. |
| 6,985,437 | B1 | 1/2006 | Vogel |
| 6,991,514 | B1 | 1/2006 | Meloni et al. |
| 7,035,593 | B2 | 4/2006 | Miller et al. |
| 7,043,207 | B2 | 5/2006 | Miyazaki |
| 7,049,965 | B2 | 5/2006 | Kelliher et al. |
| 7,110,756 | B2 | 9/2006 | Diener |
| 7,116,943 | B2 | 10/2006 | Sugar et al. |
| 7,146,176 | B2 | 12/2006 | Mchenry |
| 7,151,790 | B1 | 12/2006 | Patenaude et al. |
| 7,151,938 | B2 | 12/2006 | Weigand |
| 7,152,025 | B2 | 12/2006 | Lusky et al. |
| 7,162,207 | B2 | 1/2007 | Kursula et al. |
| 7,171,161 | B2 | 1/2007 | Miller |
| 7,187,326 | B2 | 3/2007 | Beadle et al. |
| 7,206,350 | B2 | 4/2007 | Korobkov et al. |
| 7,215,716 | B1 | 5/2007 | Smith |
| 7,254,191 | B2 | 8/2007 | Sugar et al. |
| 7,269,151 | B2 | 9/2007 | Diener et al. |
| 7,289,733 | B1 | 10/2007 | He |
| 7,292,656 | B2 | 11/2007 | Kloper et al. |

(56)

References Cited

U.S. PATENT DOCUMENTS

7,298,327	B2	11/2007	Dupray et al.	8,290,503	B2	10/2012	Sadek et al.
7,340,375	B1	3/2008	Patenaud et al.	8,295,859	B1	10/2012	Yarkan et al.
7,366,463	B1	4/2008	Archer et al.	8,295,877	B2	10/2012	Hui et al.
7,408,907	B2	8/2008	Diener	8,301,075	B2	10/2012	Sherman et al.
7,424,268	B2	9/2008	Diener et al.	8,305,215	B2	11/2012	Markhovsky et al.
7,428,270	B1	9/2008	Dubuc et al.	8,311,483	B2	11/2012	Tillman et al.
7,430,254	B1	9/2008	Anderson	8,311,509	B2	11/2012	Feher
7,459,898	B1	12/2008	Woodings	8,315,571	B2	11/2012	Lindoff et al.
7,466,960	B2	12/2008	Sugar	8,320,910	B2	11/2012	Bobier
7,471,683	B2	12/2008	Maher, III et al.	8,326,240	B1	12/2012	Kadambe et al.
7,522,917	B1	4/2009	Purdy, Jr. et al.	8,326,309	B2	12/2012	Mody et al.
7,555,262	B2	6/2009	Brenner	8,326,313	B2	12/2012	McHenry et al.
7,564,816	B2	7/2009	McHenry et al.	8,335,204	B2	12/2012	Samarasooriya et al.
7,595,754	B2	9/2009	Mehta	8,346,273	B2	1/2013	Weigand
7,606,335	B2	10/2009	Kloper et al.	8,350,970	B2	1/2013	Birkett et al.
7,606,597	B2	10/2009	Weigand	8,352,223	B1	1/2013	Anthony et al.
7,620,396	B2	11/2009	Floam et al.	8,358,723	B1	1/2013	Hamkins et al.
7,653,020	B2	1/2010	Roberts	8,364,188	B2	1/2013	Srinivasan et al.
7,676,192	B1	3/2010	Wilson	8,369,305	B2	2/2013	Diener et al.
7,692,532	B2	4/2010	Fischer et al.	8,373,759	B2	2/2013	Samarasooriya et al.
7,692,573	B1	4/2010	Funk	8,391,794	B2	3/2013	Sawai et al.
7,702,044	B2	4/2010	Nallapureddy et al.	8,391,796	B2	3/2013	Srinivasan et al.
7,725,110	B2	5/2010	Weigand	8,401,564	B2	3/2013	Singh
7,728,755	B1	6/2010	Jocic	8,406,776	B2	3/2013	Jallon
7,801,490	B1	9/2010	Scherzer	8,406,780	B2	3/2013	Mueck
7,813,742	B1	10/2010	Mitchell	RE44,142	E	4/2013	Wilson
7,835,319	B2	11/2010	Sugar	8,421,676	B2	4/2013	Moshfeghi
7,865,140	B2	1/2011	Levien et al.	8,422,453	B2	4/2013	Abedi
7,893,875	B1	2/2011	Smith	8,422,958	B2	4/2013	Du et al.
7,929,508	B1	4/2011	Yucek et al.	RE44,237	E	5/2013	McHenry
7,933,344	B2	4/2011	Hassan et al.	8,437,700	B2	5/2013	Mody et al.
7,945,215	B2	5/2011	Tang	8,442,445	B2	5/2013	Mody et al.
7,953,549	B2	5/2011	Graham et al.	8,447,237	B2	5/2013	Reial et al.
7,965,641	B2	6/2011	Ben Letaief et al.	8,451,751	B2	5/2013	Challapali et al.
8,001,901	B2	8/2011	Bass	8,463,195	B2	6/2013	Shellhammer
8,006,195	B1	8/2011	Woodings et al.	8,467,353	B2	6/2013	Proctor
8,023,957	B2	9/2011	Weigand	8,483,155	B1	7/2013	Banerjea et al.
8,026,846	B2	9/2011	Mcfadden et al.	8,494,464	B1	7/2013	Kadambe et al.
8,027,249	B2	9/2011	McHenry et al.	8,503,955	B2	8/2013	Kang et al.
8,027,690	B2	9/2011	Shellhammer	8,504,087	B2	8/2013	Stanforth et al.
8,045,654	B1	10/2011	Anderson	8,514,729	B2	8/2013	Blackwell
8,045,660	B1	10/2011	Gupta	8,515,473	B2	8/2013	Mody et al.
8,055,204	B2	11/2011	Livics et al.	8,520,606	B2	8/2013	Cleveland
8,059,694	B2	11/2011	Junell et al.	RE44,492	E	9/2013	McHenry
8,060,017	B2	11/2011	Schlicht et al.	8,526,974	B2	9/2013	Olsson et al.
8,060,035	B2	11/2011	Haykin	8,532,686	B2	9/2013	Schmidt et al.
8,060,104	B2	11/2011	Chaudhri et al.	8,538,339	B2	9/2013	Hu et al.
8,064,840	B2	11/2011	McHenry et al.	8,548,521	B2	10/2013	Hui et al.
8,077,662	B2	12/2011	Srinivasan et al.	8,554,264	B1	10/2013	Gibbons et al.
RE43,066	E	1/2012	McHenry	8,559,301	B2	10/2013	McHenry et al.
8,094,610	B2	1/2012	Wang et al.	8,565,811	B2	10/2013	Tan et al.
8,107,391	B2	1/2012	Wu et al.	8,599,024	B2	12/2013	Bloy
8,125,213	B2	2/2012	Goguillon et al.	8,718,838	B2	5/2014	Kokkeby et al.
8,131,239	B1	3/2012	Walker et al.	8,761,051	B2	6/2014	Brisebois et al.
8,134,493	B2	3/2012	Noble et al.	8,773,966	B1	7/2014	Petrovic et al.
8,151,311	B2	4/2012	Huffman et al.	8,780,968	B1	7/2014	Garcia et al.
8,155,039	B2	4/2012	Wu et al.	8,798,548	B1	8/2014	Carbajal
8,155,649	B2	4/2012	McHenry et al.	8,805,291	B1	8/2014	Garcia et al.
8,160,839	B1	4/2012	Woodings et al.	8,818,283	B2	8/2014	McHenry et al.
8,170,577	B2	5/2012	Singh	8,824,536	B1	9/2014	Garcia et al.
8,175,539	B2	5/2012	Diener et al.	8,843,155	B2	9/2014	Burton et al.
8,184,653	B2	5/2012	Dain et al.	8,886,210	B1	11/2014	Sugar
8,193,981	B1	6/2012	Hwang et al.	8,941,491	B2	1/2015	Polk et al.
8,213,868	B2	7/2012	Du et al.	8,977,212	B2	3/2015	Carbajal
8,224,254	B2	7/2012	Haykin	9,007,262	B1	4/2015	Witzgall
8,229,368	B1	7/2012	Immendorf et al.	9,008,587	B2	4/2015	Carbajal
8,233,928	B2	7/2012	Stanforth et al.	9,078,162	B2	7/2015	Garcia et al.
8,238,247	B2	8/2012	Wu et al.	9,143,968	B1	9/2015	Manku et al.
8,249,028	B2	8/2012	Porras et al.	9,185,591	B2	11/2015	Carbajal
8,249,631	B2	8/2012	Sawai	9,229,102	B1	1/2016	Wright et al.
8,260,207	B2	9/2012	Srinivasan et al.	9,245,378	B1	1/2016	Villagomez et al.
8,265,684	B2	9/2012	Sawai	9,288,683	B2	3/2016	Garcia et al.
8,279,786	B1	10/2012	Smith et al.	9,356,727	B2	5/2016	Immendorf et al.
8,280,433	B2	10/2012	Quinn et al.	9,397,619	B2	7/2016	Lozhkin
8,289,907	B2	10/2012	Seidel et al.	9,412,278	B1	8/2016	Gong et al.
				9,413,574	B1	8/2016	Timofeev et al.
				9,414,237	B2	8/2016	Garcia et al.
				9,439,078	B2	9/2016	Menon et al.
				9,466,881	B1	10/2016	Berry et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,529,360 B1	12/2016	Melamed et al.	12,183,213 B1	12/2024	Kleinbeck et al.
9,537,586 B2	1/2017	Carbajal	12,184,963 B2	12/2024	Kleinbeck
9,538,040 B2	1/2017	Goergen et al.	12,243,431 B2	3/2025	Kleinbeck et al.
9,572,055 B2	2/2017	Immendorf et al.	2001/0000959 A1	5/2001	Campana et al.
9,635,669 B2	4/2017	Gormley et al.	2001/0005423 A1	6/2001	Rhoads
9,658,341 B2	5/2017	Mathews et al.	2001/0016503 A1	8/2001	Kang
9,674,684 B1	6/2017	Mendelson	2001/0020220 A1	9/2001	Kurosawa
9,674,836 B2	6/2017	Gormley et al.	2002/0044082 A1	4/2002	Woodington et al.
9,686,789 B2	6/2017	Gormley et al.	2002/0070889 A1	6/2002	Griffin et al.
9,715,009 B1	7/2017	Parker et al.	2002/0072331 A1	6/2002	Fischer et al.
9,749,069 B2	8/2017	Garcia et al.	2002/0097184 A1	7/2002	Mayersak
9,755,972 B1	9/2017	Mao et al.	2002/0119754 A1	8/2002	Wakutsu et al.
9,767,699 B1	9/2017	Borghese et al.	2002/0161775 A1	10/2002	Lasensky et al.
9,769,834 B2	9/2017	Immendorf et al.	2002/0173341 A1	11/2002	Abdelmonem et al.
9,805,273 B1	10/2017	Seeber et al.	2003/0013454 A1	1/2003	Hunzinger
9,819,441 B2	11/2017	Immendorf et al.	2003/0040277 A1	2/2003	Deats
9,858,947 B2	1/2018	Hearing et al.	2003/0083091 A1	5/2003	Nuutinen et al.
9,862,489 B1	1/2018	Weinstein et al.	2003/0087648 A1	5/2003	Mezhvinsky et al.
9,923,700 B2	3/2018	Gormley et al.	2003/0104831 A1	6/2003	Razavilar et al.
9,942,775 B2	4/2018	Yun et al.	2003/0144601 A1	7/2003	Pritchep
9,973,278 B2	5/2018	Wang et al.	2003/0145008 A1	7/2003	Burrell
9,989,633 B1	6/2018	Pandey et al.	2003/0145328 A1	7/2003	Rabinowitz et al.
9,998,243 B2	6/2018	Garcia et al.	2003/0198304 A1	10/2003	Sugar et al.
10,027,429 B1	7/2018	Kiannejad	2003/0206640 A1	11/2003	Malvar et al.
10,104,559 B2	10/2018	Immendorf et al.	2003/0224801 A1	12/2003	Lovberg et al.
10,157,548 B2	12/2018	Priest	2003/0232612 A1	12/2003	Richards et al.
10,194,324 B2	1/2019	Yun et al.	2004/0001688 A1	1/2004	Shen
10,198,955 B1	2/2019	Boyd et al.	2004/0023674 A1	2/2004	Miller
10,227,429 B2	3/2019	Watanabe et al.	2004/0127214 A1	7/2004	Reddy et al.
10,235,523 B1	3/2019	Keller, III et al.	2004/0137915 A1	7/2004	Diener et al.
10,241,140 B2	3/2019	Moinuddin	2004/0147254 A1	7/2004	Reddy et al.
10,251,242 B1	4/2019	Rosen et al.	2004/0171390 A1	9/2004	Chitrapu
10,281,570 B2	5/2019	Parker et al.	2004/0203725 A1	10/2004	Lahav et al.
10,389,616 B2	8/2019	Ryan et al.	2004/0203826 A1	10/2004	Sugar et al.
10,393,784 B2	8/2019	Logan et al.	2004/0208238 A1	10/2004	Thomas et al.
10,408,936 B2	9/2019	Van Voorst	2004/0219885 A1	11/2004	Sugar et al.
10,459,020 B2	10/2019	Dzierwa et al.	2004/0233100 A1	11/2004	Dibble et al.
10,540,905 B2	1/2020	Bohanan et al.	2005/0003828 A1	1/2005	Sugar et al.
10,552,738 B2	2/2020	Holt et al.	2005/0096026 A1	5/2005	Chitrapu et al.
10,587,352 B2	3/2020	Kiannejad	2005/0107102 A1	5/2005	Yoon et al.
10,594,034 B1	3/2020	Tran et al.	2005/0114023 A1	5/2005	Williamson et al.
10,613,209 B2	4/2020	Emami et al.	2005/0152317 A1	7/2005	Awater et al.
10,642,813 B1	5/2020	Lazier et al.	2005/0159928 A1	7/2005	Moser
10,698,076 B2	6/2020	Jones et al.	2005/0176401 A1	8/2005	Nanda et al.
10,700,721 B2	6/2020	Ayala et al.	2005/0185618 A1	8/2005	Friday et al.
10,701,574 B2	6/2020	Gormley et al.	2005/0192727 A1	9/2005	Shostak et al.
10,764,718 B1	9/2020	Boettcher et al.	2005/0227625 A1	10/2005	Diener
10,784,974 B2	9/2020	Menon	2005/0285792 A1	12/2005	Sugar et al.
10,811,771 B1	10/2020	Tran et al.	2006/0025118 A1	2/2006	Chitrapu et al.
10,907,940 B1	2/2021	Parker et al.	2006/0047704 A1	3/2006	Gopalakrishnan
10,916,845 B2	2/2021	Tran et al.	2006/0074558 A1	4/2006	Williamson et al.
10,917,797 B2	2/2021	Menon et al.	2006/0080040 A1	4/2006	Garczarek et al.
11,012,340 B2	5/2021	Ryan et al.	2006/0111899 A1	5/2006	Padhi et al.
11,035,929 B2	6/2021	Parker et al.	2006/0128311 A1	6/2006	Tesfai
11,063,653 B2	7/2021	Ottersten et al.	2006/0133263 A1	6/2006	Bernard et al.
11,190,233 B2	11/2021	Lo et al.	2006/0199546 A1	9/2006	Durgin
11,223,431 B2	1/2022	Garcia et al.	2006/0235574 A1	10/2006	Lapinski et al.
11,265,652 B2	3/2022	Kallai et al.	2006/0238417 A1	10/2006	Jendbro et al.
11,321,282 B2	5/2022	Tran	2006/0258347 A1	11/2006	Chitrapu
11,334,807 B1	5/2022	O'Shea et al.	2007/0003059 A1*	1/2007	Langelaar
11,336,011 B2	5/2022	Tran et al.			G06T 1/0028
11,516,071 B2	11/2022	Karapantelakis et al.			380/241
11,637,641 B1	4/2023	Garcia et al.	2007/0016412 A1	1/2007	Mehrotra et al.
11,663,992 B2	5/2023	Canberk et al.	2007/0041481 A1	2/2007	Malkemes et al.
11,671,839 B2	6/2023	Guo et al.	2007/0049823 A1	3/2007	Li
11,700,304 B2	7/2023	Brown, Jr. et al.	2007/0076657 A1	4/2007	Woodings et al.
11,757,185 B2	9/2023	Tran et al.	2007/0098089 A1	5/2007	Li et al.
11,777,783 B2	10/2023	Meirosu et al.	2007/0111746 A1	5/2007	Anderson
11,791,913 B2	10/2023	Garcia et al.	2007/0149216 A1	6/2007	Misikangas
11,871,103 B2	1/2024	Kleinbeck	2007/0168580 A1	7/2007	Schumacher
11,880,888 B1	1/2024	Gold et al.	2007/0171889 A1	7/2007	Kwon et al.
11,889,351 B2	1/2024	Tagg	2007/0203645 A1	8/2007	Dees et al.
11,910,305 B2	2/2024	Buyukdura	2007/0223419 A1	9/2007	Ji et al.
12,095,518 B2	9/2024	Garcia et al.	2007/0233336 A1	10/2007	Serguei et al.
12,156,037 B2	11/2024	Montalvo	2007/0233409 A1	10/2007	Boyan et al.
			2007/0273581 A1	11/2007	Garrison et al.
			2007/0293171 A1	12/2007	Li et al.
			2007/0296591 A1	12/2007	Frederick et al.
			2007/0297541 A1	12/2007	McGehee
			2008/0001735 A1	1/2008	Tran

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0010040	A1	1/2008	McGehee	2011/0116484	A1	5/2011	Henry
2008/0090563	A1	4/2008	Chitrapu	2011/0117869	A1	5/2011	Woodings
2008/0113634	A1	5/2008	Gates et al.	2011/0122855	A1	5/2011	Henry
2008/0123731	A1	5/2008	Wegener	2011/0129006	A1	6/2011	Jung et al.
2008/0129367	A1	6/2008	Murata et al.	2011/0131260	A1	6/2011	Mody
2008/0130519	A1	6/2008	Bahl et al.	2011/0134878	A1	6/2011	Geiger et al.
2008/0133190	A1	6/2008	Peretz et al.	2011/0136478	A1	6/2011	Trigui
2008/0180325	A1	7/2008	Chung et al.	2011/0151876	A1	6/2011	Ishii et al.
2008/0186235	A1	8/2008	Struckman et al.	2011/0183621	A1	7/2011	Quan et al.
2008/0195584	A1	8/2008	Nath et al.	2011/0183685	A1	7/2011	Burton et al.
2008/0209117	A1	8/2008	Kajigaya	2011/0185059	A1	7/2011	Adnani et al.
2008/0211481	A1	9/2008	Chen	2011/0188544	A1	8/2011	Ponnuswamy
2008/0214903	A1	9/2008	Orbach	2011/0235728	A1	9/2011	Karabinis
2008/0252516	A1	10/2008	Ho et al.	2011/0237243	A1	9/2011	Guvenc et al.
2008/0261509	A1	10/2008	Sen	2011/0241923	A1	10/2011	Chernukhin
2008/0293353	A1	11/2008	Mody et al.	2011/0273328	A1	11/2011	Parker
2009/0006103	A1	1/2009	Koishida et al.	2011/0286555	A1	11/2011	Cho et al.
2009/0011713	A1	1/2009	Abusubaih et al.	2011/0286604	A1	11/2011	Matsuo
2009/0018422	A1	1/2009	Banet et al.	2011/0287779	A1	11/2011	Harper
2009/0021420	A1	1/2009	Sahinoglu	2011/0299481	A1	12/2011	Kim et al.
2009/0046003	A1	2/2009	Tung et al.	2011/0300849	A1	12/2011	Chan
2009/0046625	A1	2/2009	Diener et al.	2011/0319120	A1	12/2011	Chen et al.
2009/0066578	A1	3/2009	Beadle et al.	2012/0014332	A1	1/2012	Smith et al.
2009/0086993	A1	4/2009	Kawaguchi et al.	2012/0032854	A1	2/2012	Bull et al.
2009/0103094	A1	4/2009	Hilfiker et al.	2012/0039284	A1	2/2012	Barbieri et al.
2009/0111463	A1	4/2009	Simms et al.	2012/0040602	A1	2/2012	Charland
2009/0131067	A1	5/2009	Aaron	2012/0047544	A1	2/2012	Bouchard
2009/0135046	A1	5/2009	Steele et al.	2012/0052869	A1	3/2012	Lindoff et al.
2009/0136052	A1	5/2009	Hohlfeld et al.	2012/0058775	A1	3/2012	Dupray et al.
2009/0143019	A1	6/2009	Shellhammer	2012/0063302	A1	3/2012	Damnjanovic et al.
2009/0146881	A1	6/2009	Mesecher	2012/0071188	A1	3/2012	Wang et al.
2009/0149202	A1	6/2009	Hill et al.	2012/0072986	A1	3/2012	Livics et al.
2009/0190511	A1	7/2009	Li et al.	2012/0077510	A1	3/2012	Chen et al.
2009/0207950	A1	8/2009	Tsuruta et al.	2012/0078071	A1	3/2012	Bohm et al.
2009/0224957	A1	9/2009	Chung et al.	2012/0081248	A1	4/2012	Kennedy et al.
2009/0245327	A1	10/2009	Michaels	2012/0094681	A1	4/2012	Freda et al.
2009/0278733	A1	11/2009	Haworth	2012/0100810	A1	4/2012	Oksanen et al.
2009/0280748	A1	11/2009	Shan et al.	2012/0105066	A1	5/2012	Marvin et al.
2009/0282130	A1	11/2009	Antoniou et al.	2012/0115522	A1	5/2012	Nama et al.
2009/0285173	A1	11/2009	Koorapaty et al.	2012/0115525	A1	5/2012	Kang et al.
2009/0286563	A1	11/2009	Ji et al.	2012/0120892	A1	5/2012	Freda et al.
2009/0322510	A1	12/2009	Berger et al.	2012/0129522	A1	5/2012	Kim et al.
2010/0020707	A1	1/2010	Woodings	2012/0140236	A1	6/2012	Babbitt et al.
2010/0044122	A1	2/2010	Sleeman et al.	2012/0142386	A1	6/2012	Mody et al.
2010/0056200	A1	3/2010	Tolonen	2012/0148068	A1	6/2012	Chandra et al.
2010/0075704	A1	3/2010	Mchenry et al.	2012/0148069	A1	6/2012	Bai et al.
2010/0109936	A1	5/2010	Levy	2012/0155217	A1	6/2012	Dellinger et al.
2010/0142454	A1	6/2010	Chang	2012/0169424	A1	7/2012	Pinarello et al.
2010/0150122	A1	6/2010	Berger et al.	2012/0182430	A1	7/2012	Birkett et al.
2010/0172443	A1	7/2010	Shim et al.	2012/0195269	A1	8/2012	Kang et al.
2010/0173586	A1	7/2010	Mchenry et al.	2012/0212628	A1	8/2012	Wu et al.
2010/0176988	A1	7/2010	Maezawa et al.	2012/0214511	A1	8/2012	Vartanian et al.
2010/0177710	A1	7/2010	Gutkin et al.	2012/0230214	A1	9/2012	Kozisek et al.
2010/0220011	A1	9/2010	Heuser	2012/0246392	A1	9/2012	Cheon
2010/0253512	A1	10/2010	Wagner et al.	2012/0264388	A1	10/2012	Guo et al.
2010/0255794	A1	10/2010	Agnew	2012/0264445	A1	10/2012	Lee et al.
2010/0255801	A1	10/2010	Gunasekara et al.	2012/0275354	A1	11/2012	Villain
2010/0259998	A1	10/2010	Kwon et al.	2012/0281000	A1	11/2012	Woodings
2010/0279680	A1	11/2010	Reudink	2012/0282942	A1	11/2012	Uusitalo et al.
2010/0292930	A1	11/2010	Koster et al.	2012/0295575	A1	11/2012	Nam
2010/0306249	A1	12/2010	Hill et al.	2012/0302190	A1	11/2012	Mchenry
2010/0309317	A1	12/2010	Wu et al.	2012/0302263	A1	11/2012	Tinnakornsriruphap et al.
2010/0325621	A1	12/2010	Andrade et al.	2012/0309288	A1	12/2012	Lu
2011/0022342	A1	1/2011	Pandharipande et al.	2012/0321024	A1	12/2012	Wasiewicz et al.
2011/0045781	A1	2/2011	Shellhammer et al.	2012/0322487	A1	12/2012	Stanforth
2011/0053604	A1	3/2011	Kim et al.	2013/0005240	A1	1/2013	Novak et al.
2011/0059747	A1	3/2011	Lindoff et al.	2013/0005374	A1	1/2013	Uusitalo et al.
2011/0070885	A1	3/2011	Ruuska et al.	2013/0012134	A1	1/2013	Jin et al.
2011/0074631	A1	3/2011	Parker	2013/0017794	A1	1/2013	Kloper et al.
2011/0077017	A1	3/2011	Yu et al.	2013/0023285	A1	1/2013	Markhovsky et al.
2011/0087639	A1	4/2011	Gurney	2013/0028111	A1	1/2013	Dain et al.
2011/0090939	A1	4/2011	Diener et al.	2013/0035108	A1	2/2013	Joslyn et al.
2011/0096770	A1	4/2011	Henry	2013/0035128	A1	2/2013	Chan et al.
2011/0102258	A1	5/2011	Underbrink et al.	2013/0045754	A1	2/2013	Markhovsky et al.
2011/0111751	A1	5/2011	Markhovsky et al.	2013/0052939	A1	2/2013	Anniballi et al.
				2013/0053054	A1	2/2013	Lovitt et al.
				2013/0062334	A1	3/2013	Bilchinsky et al.
				2013/0064197	A1	3/2013	Novak et al.
				2013/0064328	A1	3/2013	Adnani et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2013/0070639	A1	3/2013	Demura et al.	2015/0201385	A1	7/2015	Mercer et al.
2013/0090071	A1	4/2013	Abraham et al.	2015/0208312	A1	7/2015	Fudaba
2013/0095843	A1	4/2013	Smith et al.	2015/0215794	A1	7/2015	Gormley et al.
2013/0100154	A1	4/2013	Woodings et al.	2015/0215949	A1	7/2015	Gormley et al.
2013/0103684	A1	4/2013	Yee et al.	2015/0242384	A1	8/2015	Reiter
2013/0113659	A1	5/2013	Morgan	2015/0248047	A1	9/2015	Chakraborty
2013/0165051	A9	6/2013	Li et al.	2015/0289254	A1	10/2015	Garcia et al.
2013/0165134	A1	6/2013	Touag et al.	2015/0289265	A1	10/2015	Gormley et al.
2013/0165170	A1	6/2013	Kang	2015/0296386	A1	10/2015	Menon et al.
2013/0183989	A1	7/2013	Hasegawa et al.	2015/0301167	A1	10/2015	Sentelle et al.
2013/0183994	A1	7/2013	Ringstroem et al.	2015/0319768	A1	11/2015	Abdelmonem et al.
2013/0184022	A1	7/2013	Schmidt	2016/0014713	A1	1/2016	Kennedy et al.
2013/0190003	A1	7/2013	Smith et al.	2016/0037550	A1	2/2016	Barabell et al.
2013/0190028	A1	7/2013	Wang et al.	2016/0050690	A1	2/2016	Yun et al.
2013/0196677	A1	8/2013	Smith et al.	2016/0061583	A1	3/2016	Ryu et al.
2013/0208587	A1	8/2013	Bala et al.	2016/0069994	A1	3/2016	Allen et al.
2013/0210457	A1	8/2013	Kummetz	2016/0072597	A1	3/2016	Carbajal
2013/0210473	A1	8/2013	Weigand	2016/0073318	A1	3/2016	Aguirre
2013/0217406	A1	8/2013	Villardi et al.	2016/0080955	A1	3/2016	Carbajal
2013/0217408	A1	8/2013	Difazio et al.	2016/0086621	A1	3/2016	Hearing et al.
2013/0217450	A1	8/2013	Kanj et al.	2016/0095188	A1	3/2016	Verberkt et al.
2013/0231121	A1	9/2013	Kwak et al.	2016/0117853	A1	4/2016	Zhong et al.
2013/0237212	A1	9/2013	Khayrallah et al.	2016/0124071	A1	5/2016	Baxley et al.
2013/0242792	A1	9/2013	Woodings	2016/0126988	A1	5/2016	Mester et al.
2013/0242934	A1	9/2013	Ueda et al.	2016/0127110	A1	5/2016	McMeekin et al.
2013/0260703	A1	10/2013	Actis et al.	2016/0127392	A1	5/2016	Baxley et al.
2013/0265198	A1	10/2013	Stroud	2016/0154406	A1	6/2016	Im et al.
2013/0272436	A1	10/2013	Makhlouf et al.	2016/0161578	A1	6/2016	Weissler et al.
2013/0279556	A1	10/2013	Seller	2016/0198471	A1	7/2016	Young et al.
2013/0288734	A1	10/2013	Mody et al.	2016/0219506	A1	7/2016	Pratt et al.
2013/0309975	A1	11/2013	Kpodzo et al.	2016/0219590	A1	7/2016	Khawer et al.
2013/0315112	A1	11/2013	Gormley et al.	2016/0225240	A1	8/2016	Voddhi et al.
2013/0329690	A1	12/2013	Kim et al.	2016/0241910	A1	8/2016	Rowe
2013/0331114	A1	12/2013	Gormley et al.	2016/0245921	A1	8/2016	Li et al.
2014/0003547	A1	1/2014	Williams et al.	2016/0252342	A1	9/2016	Feldmann et al.
2014/0015796	A1	1/2014	Philipp	2016/0323920	A1	11/2016	Carbajal
2014/0018683	A1	1/2014	Park et al.	2016/0334527	A1	11/2016	Xu et al.
2014/0024405	A1	1/2014	Qiu	2016/0345135	A1	11/2016	Garcia et al.
2014/0064723	A1	3/2014	Adles et al.	2016/0364079	A1	12/2016	Qiu et al.
2014/0066060	A1	3/2014	Ngai	2016/0366685	A1	12/2016	Gormley et al.
2014/0073261	A1	3/2014	Hassan et al.	2016/0374088	A1	12/2016	Garcia et al.
2014/0086212	A1	3/2014	Kafle et al.	2017/0024767	A1	1/2017	Johnson, Jr. et al.
2014/0128032	A1	5/2014	Muthukumar	2017/0025996	A1	1/2017	Cheung et al.
2014/0139374	A1	5/2014	Wellman et al.	2017/0039413	A1	2/2017	Nadler
2014/0163309	A1	6/2014	Bernhard et al.	2017/0048838	A1	2/2017	Chrisikos et al.
2014/0199993	A1	7/2014	Dhanda et al.	2017/0061690	A1	3/2017	Laughlin et al.
2014/0201367	A1	7/2014	Trummer et al.	2017/0064564	A1	3/2017	Yun et al.
2014/0204766	A1	7/2014	Immendorf et al.	2017/0078792	A1	3/2017	Simons
2014/0206279	A1	7/2014	Immendorf et al.	2017/0079007	A1	3/2017	Carbajal
2014/0206307	A1	7/2014	Maurer et al.	2017/0094527	A1	3/2017	Shattil et al.
2014/0206343	A1	7/2014	Immendorf et al.	2017/0118049	A1	4/2017	Miao et al.
2014/0207414	A1	7/2014	Bordow	2017/0134631	A1	5/2017	Zhao et al.
2014/0225590	A1	8/2014	Jacobs	2017/0146462	A1	5/2017	Baker et al.
2014/0256268	A1	9/2014	Olgaard	2017/0148332	A1	5/2017	Ziemba et al.
2014/0256370	A9	9/2014	Gautier et al.	2017/0148467	A1	5/2017	Franklin et al.
2014/0269374	A1	9/2014	Abdelmonem et al.	2017/0192089	A1	7/2017	Parker et al.
2014/0269376	A1	9/2014	Garcia et al.	2017/0208495	A1	7/2017	Kleinbeck et al.
2014/0274103	A1	9/2014	Steer et al.	2017/0234979	A1	8/2017	Mathews et al.
2014/0287100	A1	9/2014	Libman	2017/0237484	A1	8/2017	Heath et al.
2014/0301216	A1	10/2014	Immendorf et al.	2017/0238201	A1	8/2017	Gormley et al.
2014/0302796	A1	10/2014	Gormley et al.	2017/0238203	A1	8/2017	Dzierwa et al.
2014/0335879	A1	11/2014	Immendorf et al.	2017/0243138	A1	8/2017	Dzierwa et al.
2014/0340684	A1	11/2014	Edler et al.	2017/0243139	A1	8/2017	Dzierwa et al.
2014/0342675	A1	11/2014	Massarella et al.	2017/0248677	A1	8/2017	Mahmood et al.
2014/0348004	A1	11/2014	Ponnuswamy	2017/0248807	A1	8/2017	Jiang et al.
2014/0362934	A1	12/2014	Kumar	2017/0250766	A1	8/2017	Dzierwa et al.
2015/0016429	A1	1/2015	Menon et al.	2017/0253330	A1	9/2017	Saigh et al.
2015/0023329	A1	1/2015	Jiang et al.	2017/0261604	A1	9/2017	Van Voorst
2015/0068296	A1	3/2015	Lanza di Scalea	2017/0261613	A1	9/2017	Van Voorst
2015/0072633	A1	3/2015	Massarella et al.	2017/0261615	A1	9/2017	Ying et al.
2015/0126181	A1	5/2015	Breuer et al.	2017/0274992	A1	9/2017	Chretien
2015/0133058	A1	5/2015	Livis et al.	2017/0289840	A1	10/2017	Sung et al.
2015/0150753	A1	6/2015	Racette	2017/0290075	A1	10/2017	Carbajal et al.
2015/0156827	A1	6/2015	Ibragimov et al.	2017/0311307	A1	10/2017	Negus et al.
2015/0170145	A1	6/2015	Patel et al.	2017/0358103	A1	12/2017	Shao et al.
				2017/0366361	A1	12/2017	Afkhami et al.
				2017/0374572	A1	12/2017	Kleinbeck et al.
				2017/0374573	A1	12/2017	Kleinbeck et al.
				2018/0006730	A1	1/2018	Kuo et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2018/0014217	A1	1/2018	Kleinbeck et al.	2020/0242603	A1	7/2020	Salkintzis
2018/0024220	A1	1/2018	Massarella et al.	2020/0245167	A1	7/2020	Kleinbeck et al.
2018/0046869	A1	2/2018	Cordell et al.	2020/0260306	A1	8/2020	Kleinbeck et al.
2018/0070362	A1	3/2018	Ryan et al.	2020/0294032	A1	9/2020	Cheng et al.
2018/0074170	A1	3/2018	Ray	2020/0295855	A1	9/2020	Kleinbeck et al.
2018/0074171	A1	3/2018	Ray et al.	2020/0382961	A1	12/2020	Shattil et al.
2018/0081355	A1	3/2018	Magy et al.	2020/0388036	A1	12/2020	Skrede et al.
2018/0083721	A1	3/2018	Wada et al.	2021/0014696	A1	1/2021	Brookes
2018/0124533	A1	5/2018	Messier et al.	2021/0067974	A1	3/2021	Guo et al.
2018/0129881	A1	5/2018	Seeber et al.	2021/0082254	A1	3/2021	Givant
2018/0131445	A1	5/2018	Esman et al.	2021/0084217	A1	3/2021	Kleinbeck
2018/0143314	A1	5/2018	Pelletier et al.	2021/0211911	A1	7/2021	Kleinbeck et al.
2018/0149729	A1	5/2018	Grandin et al.	2021/0250795	A1	8/2021	Dzierwa et al.
2018/0211179	A1	7/2018	Dzierwa	2021/0255356	A1	8/2021	Vu et al.
2018/0284758	A1	10/2018	Cella et al.	2021/0280039	A1	9/2021	Kleinbeck
2018/0288620	A1	10/2018	Jayawickrama et al.	2021/0281510	A1	9/2021	Brown, Jr. et al.
2018/0294901	A1	10/2018	Garcia et al.	2021/0306022	A1	9/2021	Fernando et al.
2018/0313877	A1	11/2018	Brant et al.	2021/0360423	A1	11/2021	Dzierwa et al.
2018/0313945	A1	11/2018	Parker et al.	2021/0360450	A1	11/2021	Kleinbeck et al.
2018/0324595	A1	11/2018	Shima	2021/0360453	A1	11/2021	Kleinbeck et al.
2018/0329020	A1	11/2018	Hafizovic et al.	2021/0360454	A1	11/2021	Carbajal et al.
2018/0331863	A1	11/2018	Carbajal	2021/0409591	A1	12/2021	Kleinbeck
2019/0004518	A1	1/2019	Zhou et al.	2022/0030541	A1	1/2022	Dzierwa et al.
2019/0011534	A1	1/2019	Trotta et al.	2022/0052770	A1	2/2022	Kleinbeck et al.
2019/0018103	A1	1/2019	Qian et al.	2022/0128612	A1	4/2022	Dzierwa et al.
2019/0064130	A1	2/2019	Kanazawa et al.	2022/0131623	A1	4/2022	Garcia et al.
2019/0064223	A1	2/2019	Kincaid	2022/0150824	A1	5/2022	Kleinbeck et al.
2019/0072601	A1	3/2019	Dzierwa et al.	2022/0174525	A1	6/2022	Dzierwa et al.
2019/0074802	A1	3/2019	Geha et al.	2022/0253407	A1	8/2022	Tran
2019/0077507	A1	3/2019	Ferris et al.	2022/0262228	A1	8/2022	Kleinbeck
2019/0123428	A1	4/2019	Packer et al.	2022/0262261	A1	8/2022	Kleinbeck
2019/0180630	A1	6/2019	Kleinbeck	2022/0286997	A1	9/2022	Kleinbeck et al.
2019/0190553	A1	6/2019	Tsuji et al.	2022/0376921	A1	11/2022	Maria
2019/0191313	A1	6/2019	Dzierwa et al.	2023/0018133	A1	1/2023	Burnette et al.
2019/0200303	A1	6/2019	Nakahara	2023/0045636	A1	2/2023	Dzierwa et al.
2019/0208112	A1	7/2019	Kleinbeck	2023/0087729	A1	3/2023	Goldstein et al.
2019/0208491	A1	7/2019	Dzierwa et al.	2023/0098387	A1	3/2023	Hafeez et al.
2019/0215709	A1	7/2019	Kleinbeck et al.	2023/0105718	A1	4/2023	Carbajal
2019/0223139	A1	7/2019	Kleinbeck et al.	2023/0114804	A1	4/2023	Kleinbeck
2019/0230539	A1	7/2019	Dzierwa et al.	2023/0116761	A1	4/2023	Barry et al.
2019/0230540	A1	7/2019	Carbajal et al.	2023/0118723	A1	4/2023	Carbajal et al.
2019/0236266	A1	8/2019	Nashimoto et al.	2023/0123375	A1	4/2023	Dzierwa et al.
2019/0245722	A1	8/2019	Carbajal	2023/0126223	A1	4/2023	Kleinbeck et al.
2019/0246304	A1	8/2019	Dzierwa et al.	2023/0189010	A1	6/2023	Muhammad et al.
2019/0253160	A1	8/2019	Garcia et al.	2023/0206216	A1	6/2023	Lehmann et al.
2019/0253905	A1	8/2019	Kleinbeck et al.	2023/0209378	A1	6/2023	Kleinbeck et al.
2019/0260768	A1	8/2019	Mestha et al.	2023/0232244	A1	7/2023	Dzierwa et al.
2019/0274059	A1	9/2019	Kleinbeck et al.	2023/0252744	A1	8/2023	Miller et al.
2019/0296910	A1	9/2019	Cheung	2023/0254054	A1	8/2023	Garcia et al.
2019/0302249	A1	10/2019	High et al.	2023/0254567	A1	8/2023	Kleinbeck
2019/0342202	A1	11/2019	Ryan et al.	2023/0254702	A1	8/2023	Damnjanovic et al.
2019/0346571	A1	11/2019	Furumoto	2023/0275791	A1	8/2023	Carbajal
2019/0360783	A1	11/2019	Whittaker	2023/0276280	A1	8/2023	Kleinbeck et al.
2019/0364533	A1	11/2019	Kleinbeck et al.	2023/0300256	A1	9/2023	Zhao et al.
2020/0034620	A1	1/2020	Lutterodt	2023/0308789	A1	9/2023	Tian et al.
2020/0036459	A1	1/2020	Menon	2023/0308915	A1	9/2023	Carbajal et al.
2020/0036487	A1	1/2020	Hammond et al.	2023/0326323	A1	10/2023	Kleinbeck
2020/0043346	A1	2/2020	Vacek	2023/0334472	A1	10/2023	Pene
2020/0059800	A1	2/2020	Menon et al.	2023/0345441	A1	10/2023	Baxley
2020/0066132	A1	2/2020	Kleinbeck	2023/0349962	A1	11/2023	Dzierwa et al.
2020/0067752	A1	2/2020	DelMarco	2023/0378645	A1	11/2023	Tran
2020/0068573	A1	2/2020	Drozdz et al.	2023/0403564	A1	12/2023	Dzierwa et al.
2020/0096548	A1	3/2020	Dzierwa et al.	2024/0007204	A1	1/2024	Kleinbeck et al.
2020/0107207	A1	4/2020	Kleinbeck et al.	2024/0023054	A1	1/2024	Dzierwa et al.
2020/0120266	A1	4/2020	Kleinbeck	2024/0029572	A1	1/2024	Kleinbeck
2020/0128418	A1	4/2020	Dzierwa et al.	2024/0031042	A1	1/2024	Garcia et al.
2020/0137583	A1	4/2020	Economy et al.	2024/0032084	A1	1/2024	Hellwig
2020/0142029	A1	5/2020	Brooker et al.	2024/0089130	A1	3/2024	Wang et al.
2020/0145032	A1	5/2020	Ayala et al.	2024/0097951	A1	3/2024	Carbajal
2020/0162890	A1	5/2020	Spencer et al.	2024/0103059	A1	3/2024	Dzierwa et al.
2020/0167196	A1	5/2020	Smith et al.	2024/0114370	A1	4/2024	Kleinbeck et al.
2020/0169892	A1	5/2020	Dzierwa et al.	2024/0160791	A1	5/2024	He et al.
2020/0184832	A1	6/2020	Kleinbeck	2024/0241163	A1	7/2024	Dzierwa et al.
2020/0196269	A1	6/2020	Dzierwa et al.	2024/0242589	A1	7/2024	Kleinbeck
2020/0196270	A1	6/2020	Kleinbeck et al.	2024/0244461	A1	7/2024	Kleinbeck et al.
				2024/0267268	A1	8/2024	Carbajal
				2024/0267769	A1	8/2024	Dzierwa et al.
				2024/0275504	A1	8/2024	Kleinbeck et al.
				2024/0276261	A1	8/2024	Kleinbeck et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2024/0284214	A1	8/2024	Kleinbeck et al.
2024/0292248	A1	8/2024	Kleinbeck et al.
2024/0330927	A1	10/2024	Abdelrahman et al.
2024/0362631	A1	10/2024	Agrawal et al.
2024/0373463	A1	11/2024	Furuichi et al.
2024/0378982	A1	11/2024	Kleinbeck
2024/0381106	A1	11/2024	Montalvo et al.
2024/0386800	A1	11/2024	Kleinbeck et al.
2024/0386801	A1	11/2024	Kleinbeck et al.
2024/0388937	A1	11/2024	Murias et al.
2024/0396648	A1	11/2024	Kleinbeck et al.
2024/0430142	A1	12/2024	Carbajal
2025/0007629	A1	1/2025	Dzierwa et al.
2025/0039711	A1	1/2025	Dzierwa et al.
2025/0063405	A1	2/2025	Carbajal et al.
2025/0071584	A1	2/2025	Kleinbeck et al.

FOREIGN PATENT DOCUMENTS

CN	117196617	A	12/2023
CN	117998368	A	5/2024
KR	100248671	B1	4/2000
KR	20140041618	A	4/2014
SU	953557	A1	8/1982
WO	2012129932	A1	10/2012
WO	2018184682	A1	10/2018
WO	2023197982	A1	10/2023
WO	2023232010	A1	12/2023

OTHER PUBLICATIONS

"A Low-Cost, Near-Real-Time Two-LIAS-Based UWB Emitter Monitoring System"; Wang et al.; IEEE A&E Systems Magazine Nov. 2015 (Year: 2015).

"Joint TDOA and FDOA Estimation: A Conditional Bound and Its Use for Optimally Weighted Localization"; Yeredor et al.; IEEE Transactions on Signal Processing, vol. 59, No. 4, Apr. 2011 (Year: 2011).

"Multipath TDOA and FDOA Estimation Using the EM Algorithm"; Belanger; Apr. 27, 1993; 1993 IEEE International Conference on Acoustics, Speech, and Signal Processing (Year: 1993).

"Noise Figure", Wikipedia, located at https://en.wikipedia.org/wiki/Noise_figure (Year: 2022).

"Signal Models for TDOA/FDOA Estimation"; Fowler et al.; IEEE Transactions on Aerospace and Electronic Systems vol. 44, No. Oct. 4, 2008 (Year: 2008).

"Specific attenuation model for rain for use in prediction methods", Recommendation ITU-R p. 838-3 (Year: 2005).

Bluetooth vs Zigbee-difference between Bluetooth and Zigbee (located at <https://www.rfwireless-world.com/Terminology/Bluetooth-vs-zigbee.html>) (Year: 2012).

Boll S.F., Suppression of Acoustic Noise in Speech Using Spectral Subtraction, Apr. 1979, IEEE Transactions on Acoustics, Speech, and Signal Processing, vol. ASSP-27, No. 2, (Year: 1979).

David Eppink and Wolf Kuebler, "TIREM/SEM Handbook", Mar. 1994, IIT Research Institute, p. 1-6, located at <http://www.dtic.mil/cgi-bin/GetTRDoc?Location=U2&doc=GetTRDoc.pdf&AD=ADA296913>.

English translation of SU-953557-A1 (Year: 2024).

Gabriel Garcia and Daniel Carbajal, U.S. Appl. No. 61/789,758, Provisional Patent Application, filed Mar. 15, 2013 (Specification, Claims, and Drawings).

Gary L. Sugar, System and method for locating wireless devices in an unsynchronized wireless network, U.S. Appl. No. 60/319,737, Provisional Patent Application filed on Nov. 27, 2002, Specification including the claims, abstract, and drawings.

IEEE 100 the Authoritative Dictionary of IEEE Standards Terms. Seventh Edition. Published by Standards Information Network IEEE Press. p. 6 (Year: 2000).

International Search Report and Written Opinion dated Jun. 21, 2018 issued by the International Application Division, Korean Intellectual Property Office as International Searching Authority in connection with International Application No. PCT/US2018/014504 (21 pages).

Mehmet Ali Aygul, Ahmed Naeem, Huseyin Arslan, "Blind Signal Analysis—Wireless Communication Signals", located at <https://doi.org/10.1002/9781119764441.ch12> (Year: 2021).

Mobile Emitter Geolocation and Tracking Using TDOA and FDOA Measurements; Musicki et al.; IEEE Transactions on Signal Processing, vol. 58, No. 3, Mar. 2010 (Year: 2010).

RF and Digital Signal Processing for Software-Defined Radio, Chapter 4: High-Level Requirements and Link Budget Analysis (Year: 2009).

S. Dörner, S. Cammerer, J. Hoydis and S. t. Brink, "Deep Learning Based Communication Over the Air," in IEEE Journal of Selected Topics in Signal Processing, vol. 12, No. 1, pp. 132-143, Feb. 2018, doi: 10.1109/JSTSP.2017.2784180.

Steven W. Smith, The Scientist & Engineer's Guide to Digital Signal Processing, 1999, California Technical Publishing, San Diego, California, 2nd Edition, p. 312 (located at http://www.analog.com/media/en/technical-documentation/dsp-book/dsp_book_ch18.pdf) (Year: 1999).

T. J. O'Shea, K. Karra and T. C. Clancy, "Learning to communicate: Channel auto-encoders, domain specific regularizers, and attention," 2016 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), Limassol, Cyprus, 2016, pp. 223-228, doi: 10.1109/ISSPIT.2016.7886039.

T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," in IEEE Transactions on Cognitive Communications and Networking, vol. 3, No. 4, pp. 563-575, Dec. 2017, doi: 10.1109/TCCN.2017.2758370.

* cited by examiner

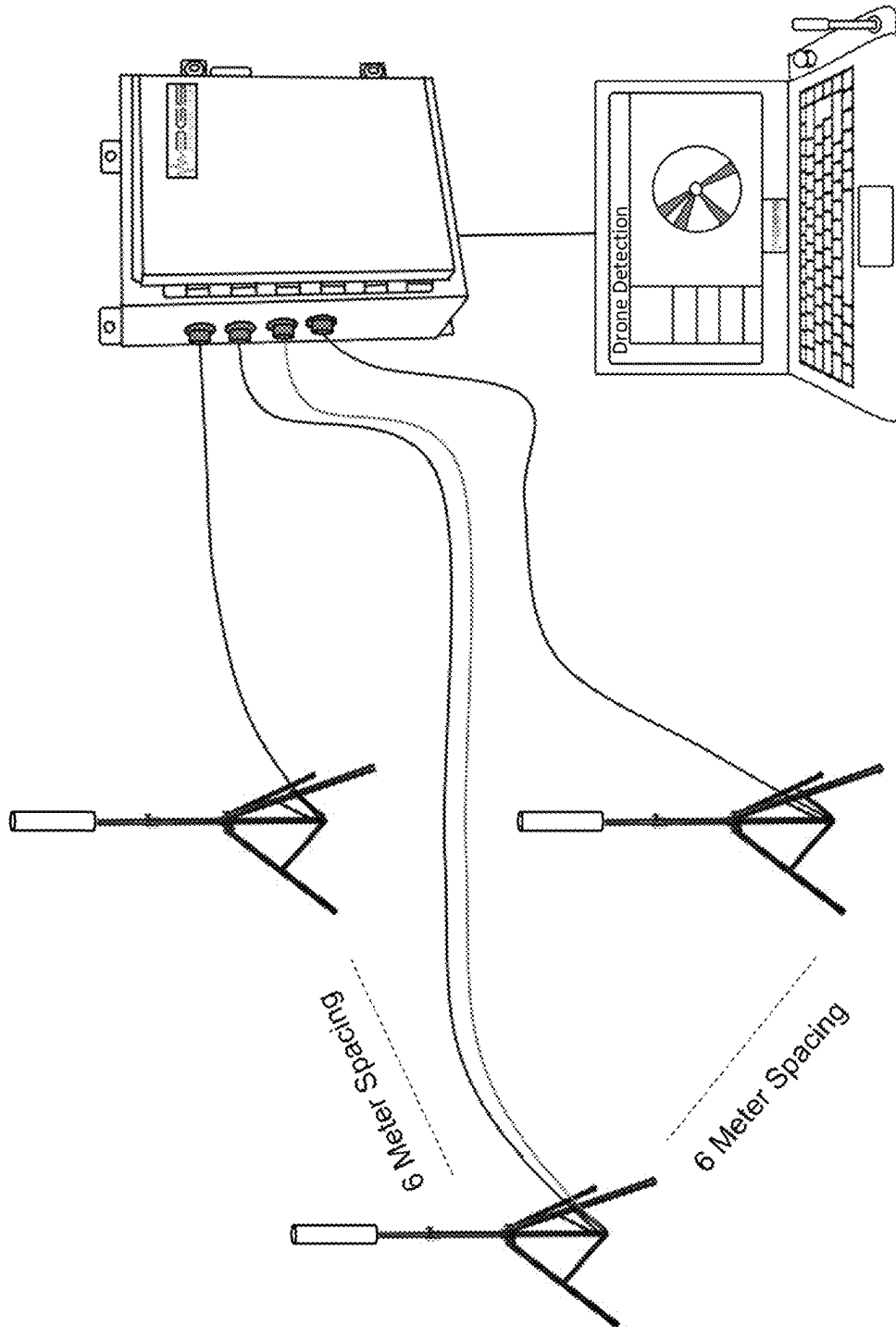


FIG. 1

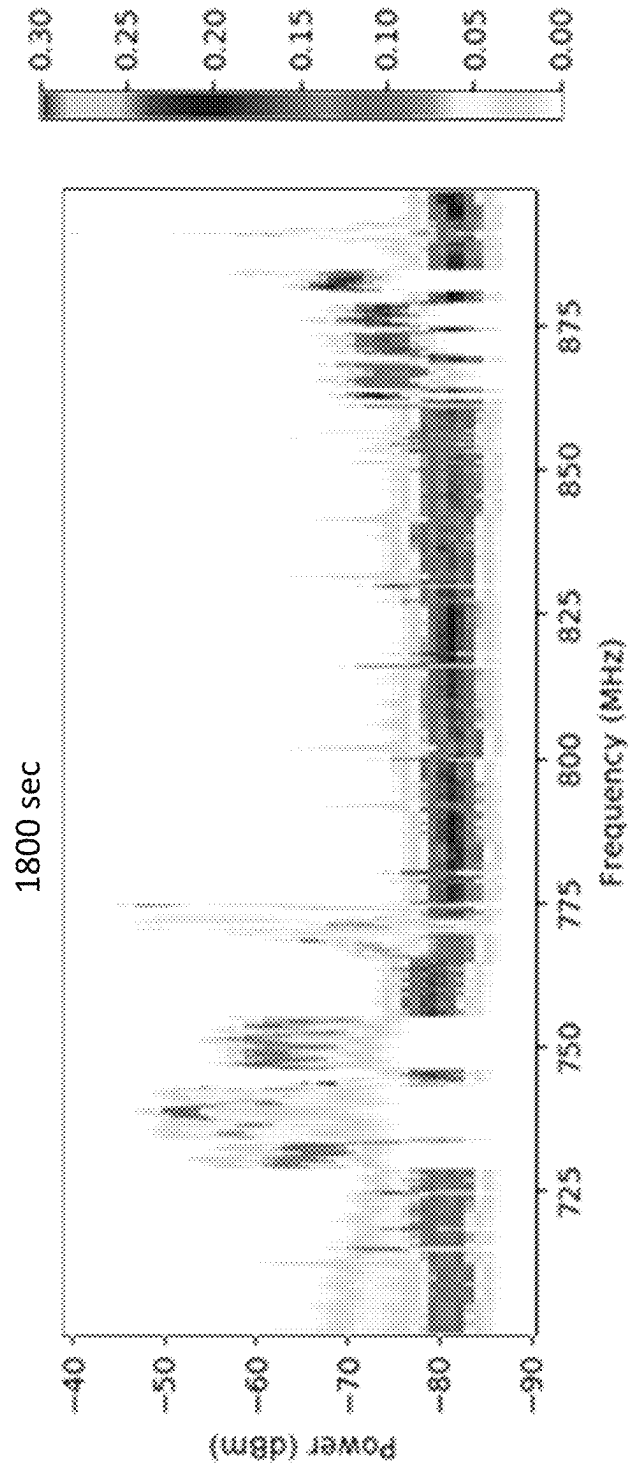


FIG. 2

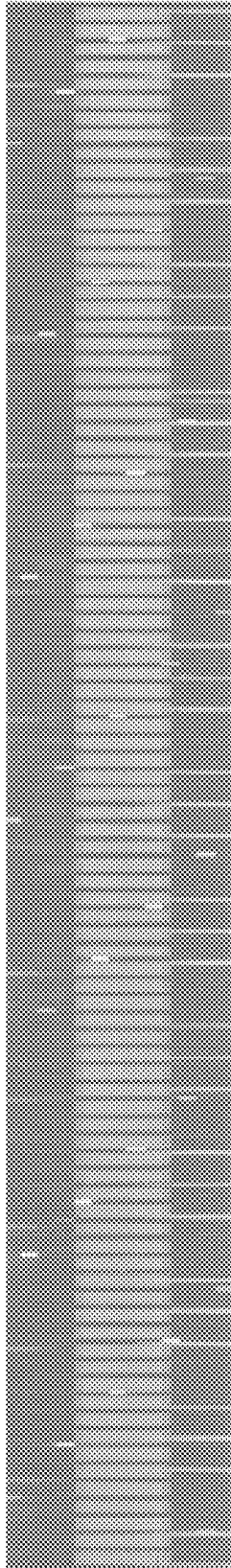


FIG. 3

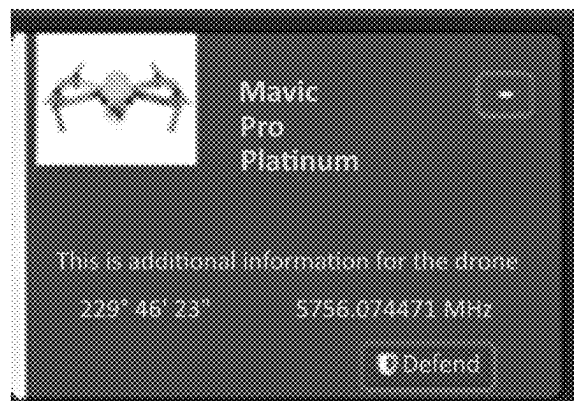
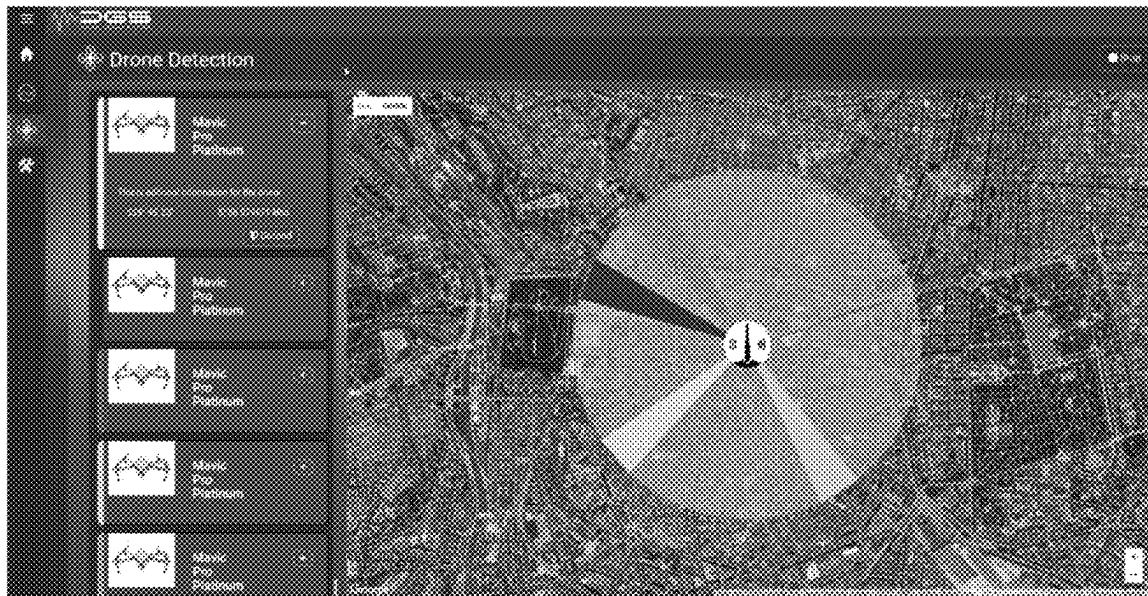


FIG. 4

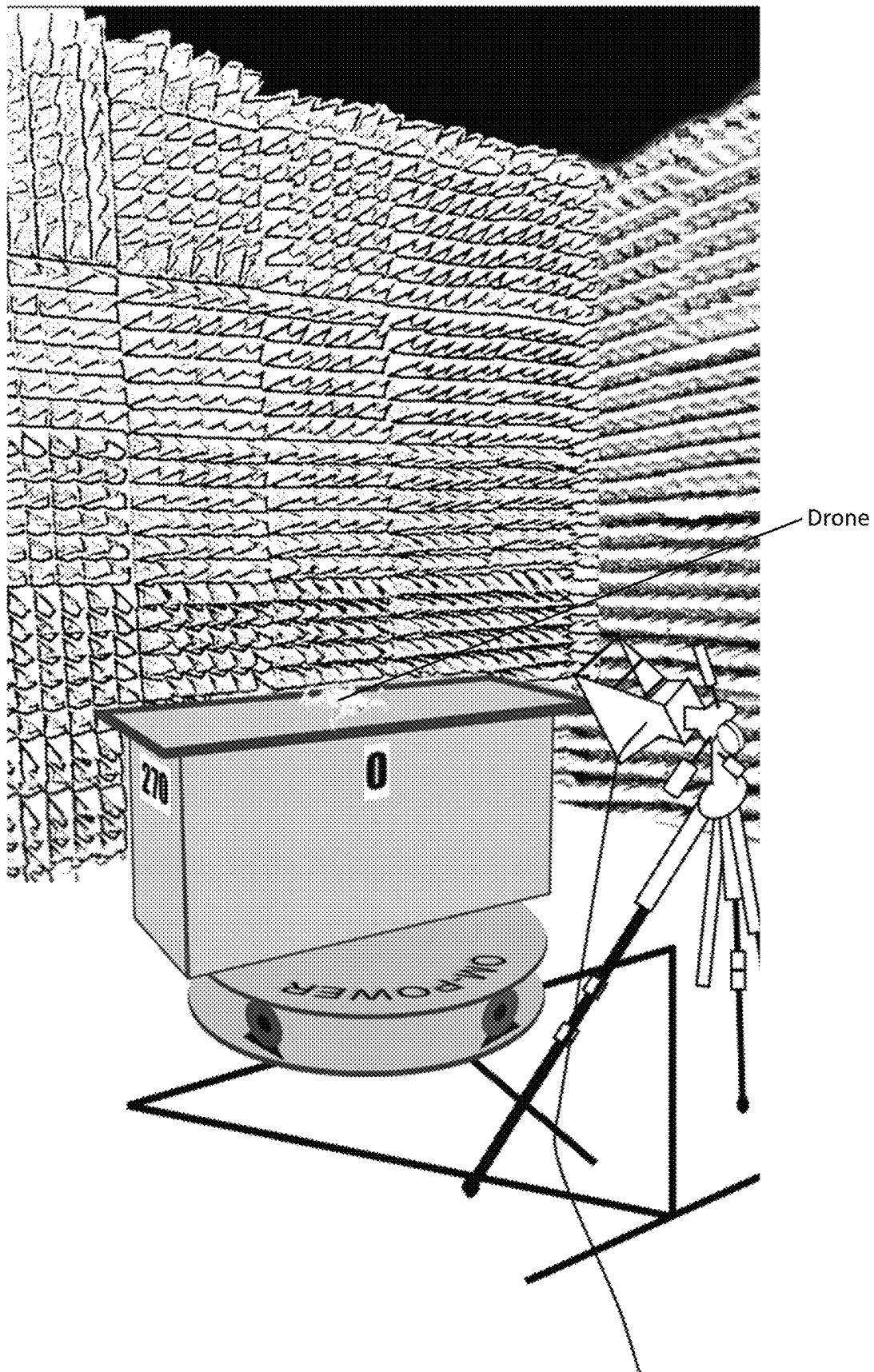


FIG. 5

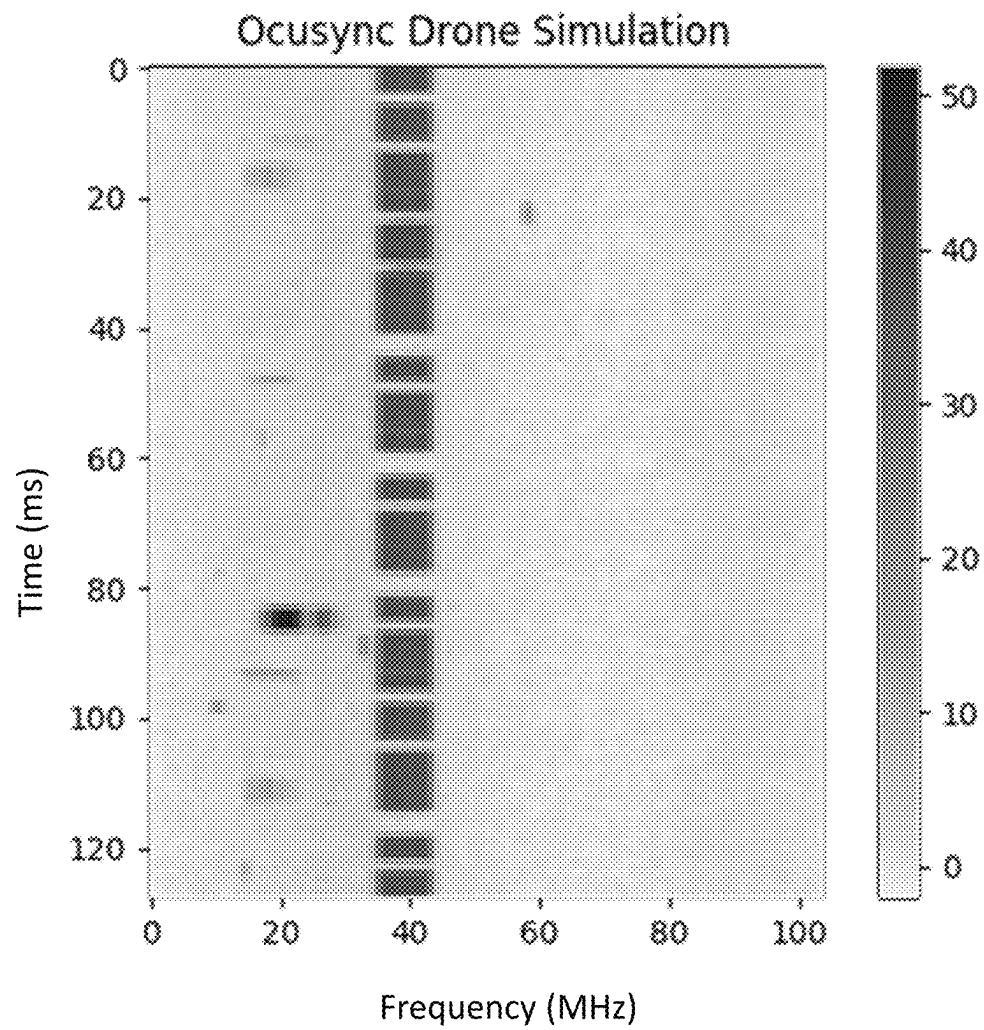


FIG. 6

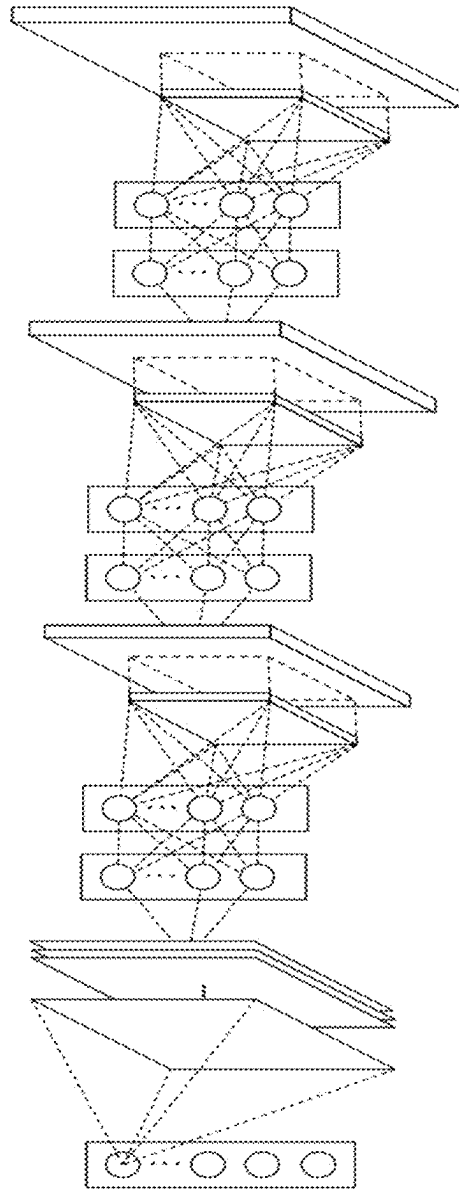


FIG. 7

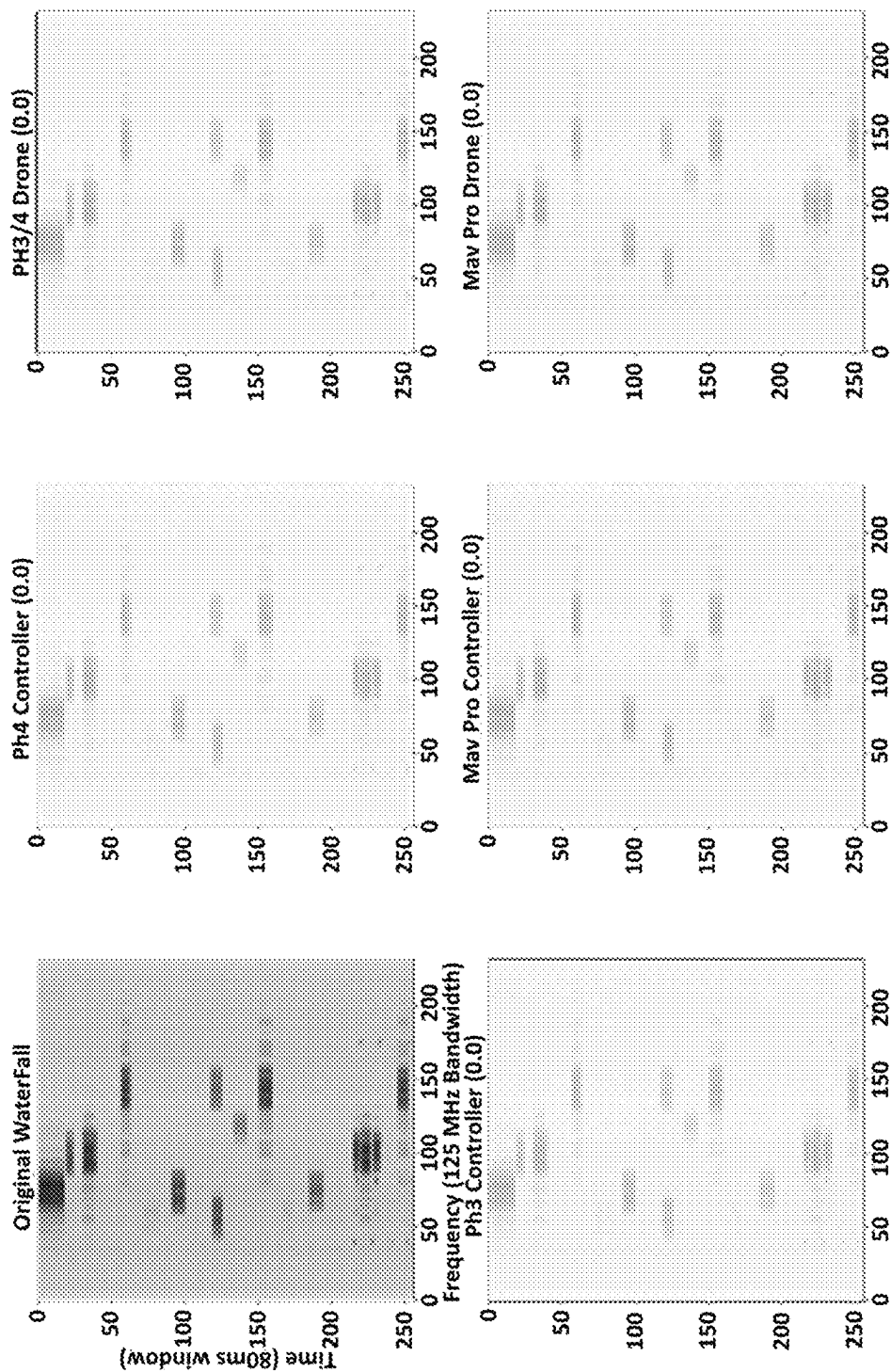


FIG. 8

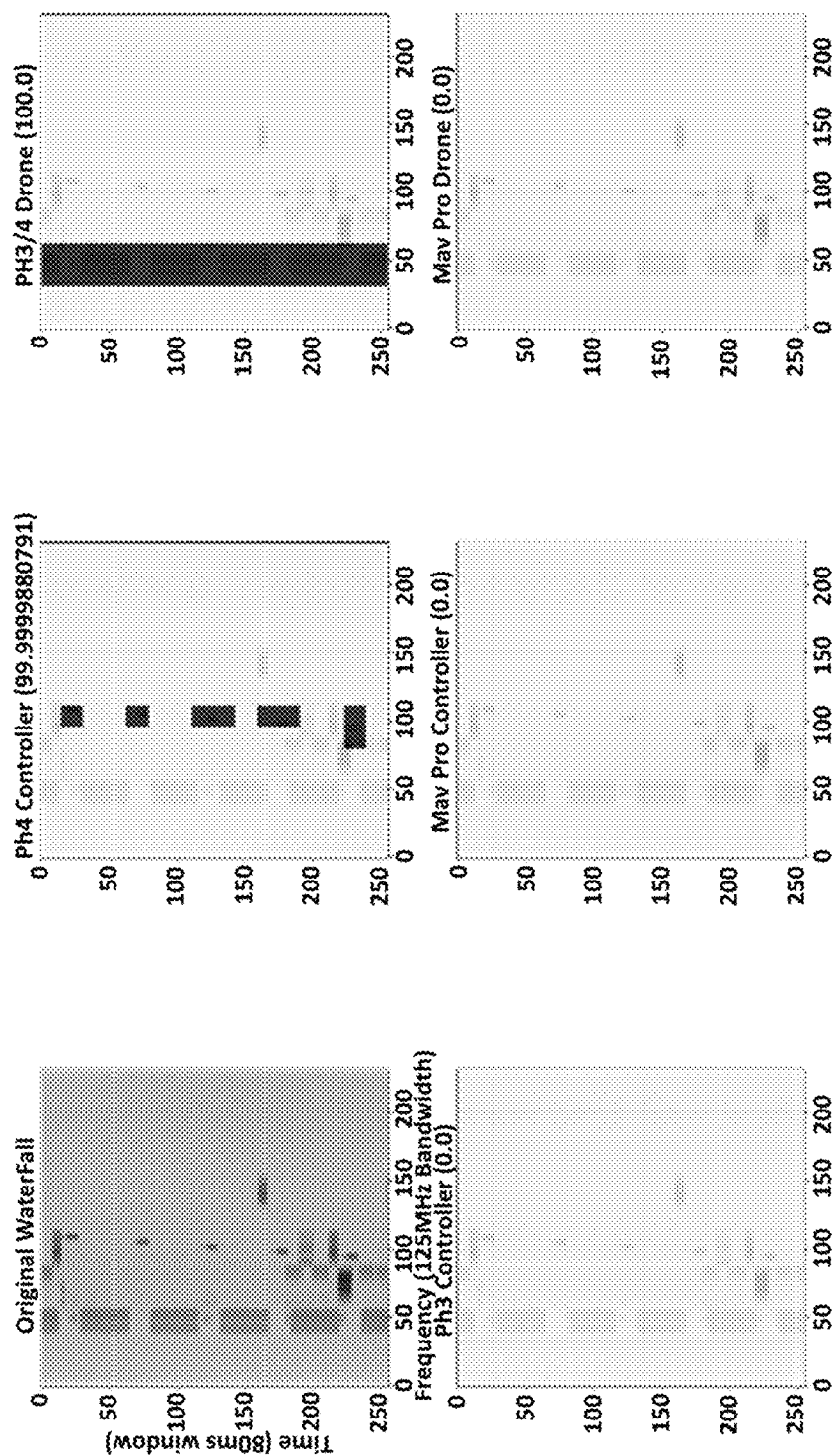


FIG. 9

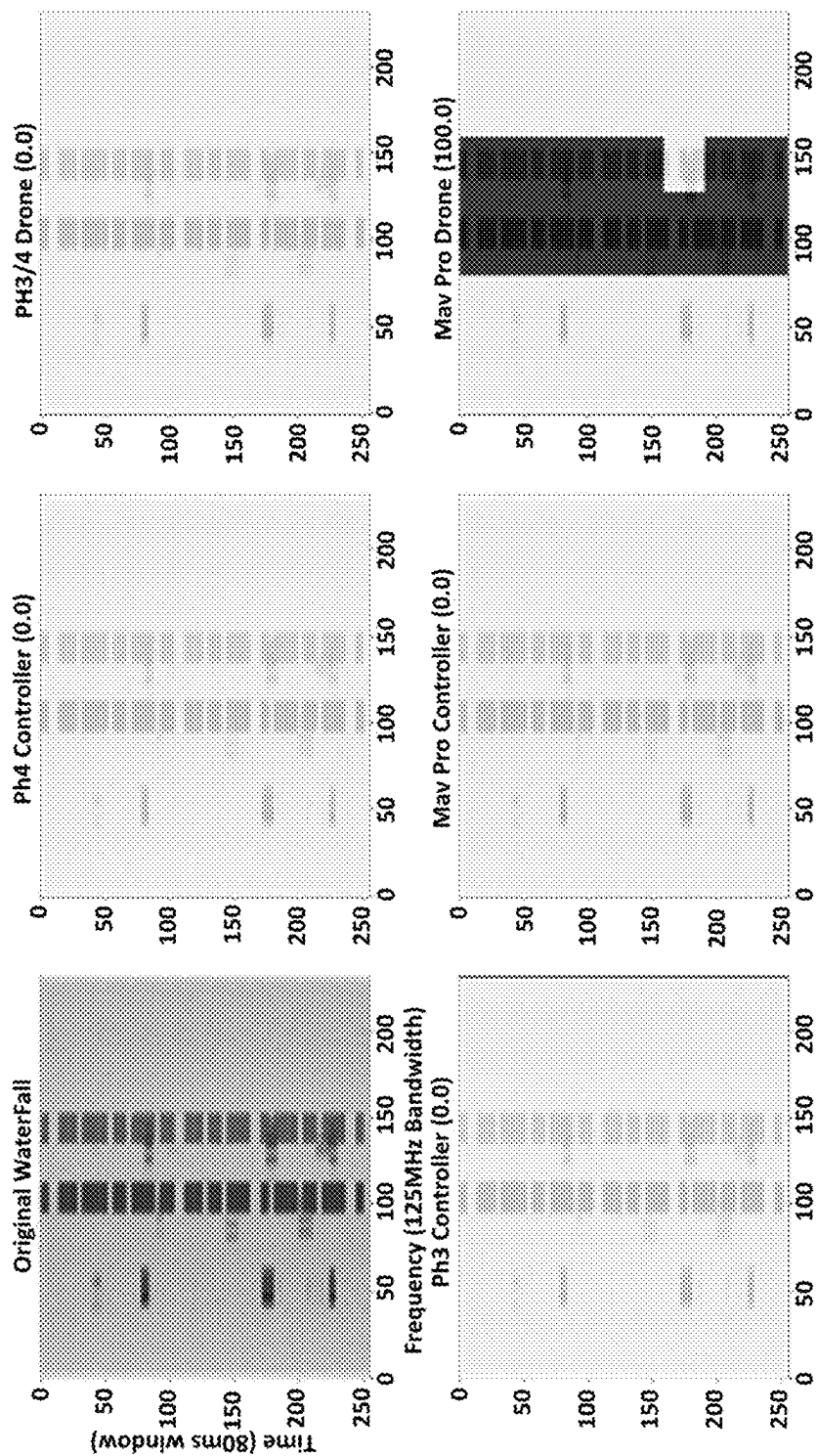


FIG. 10

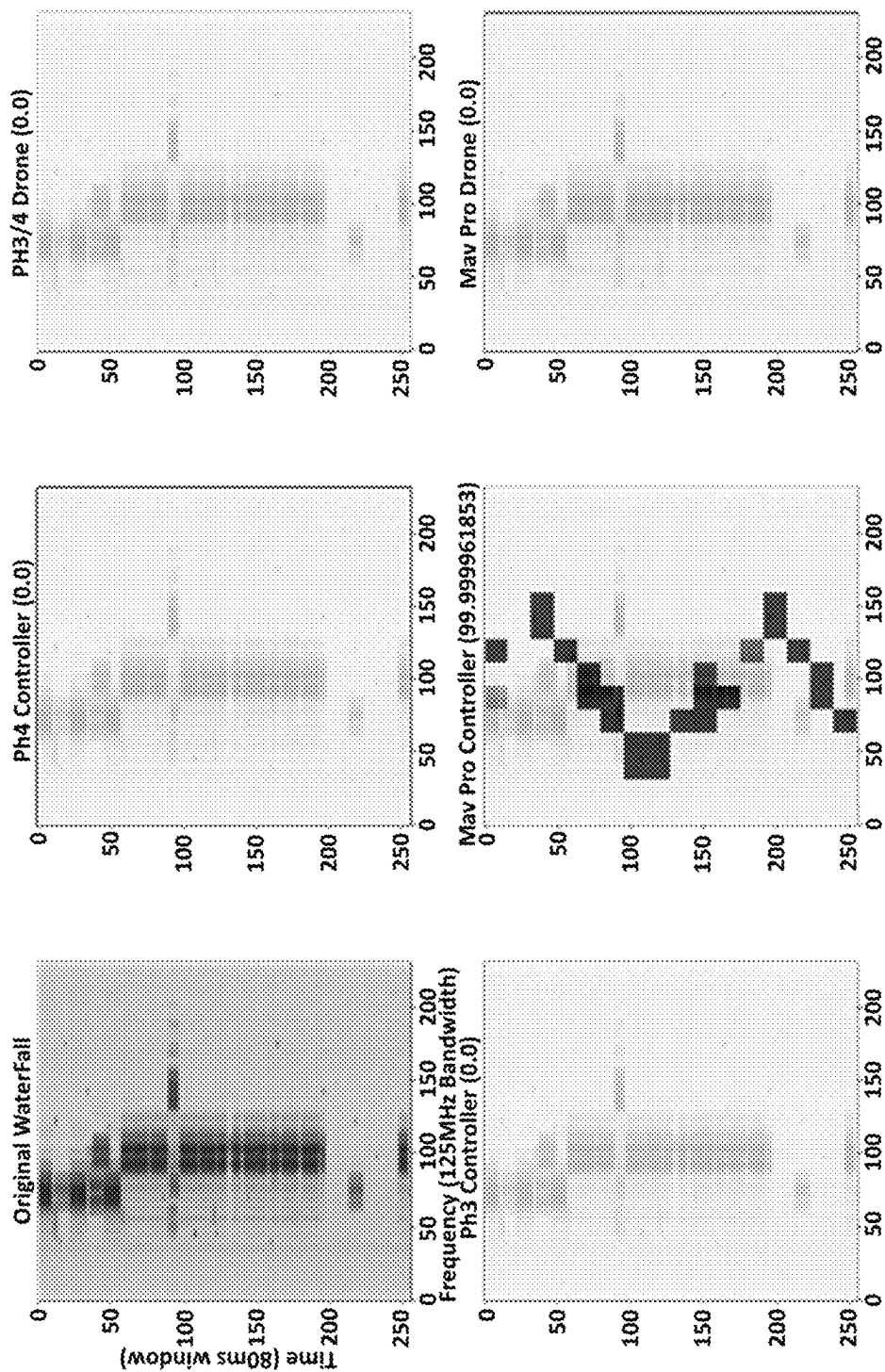


FIG. 11

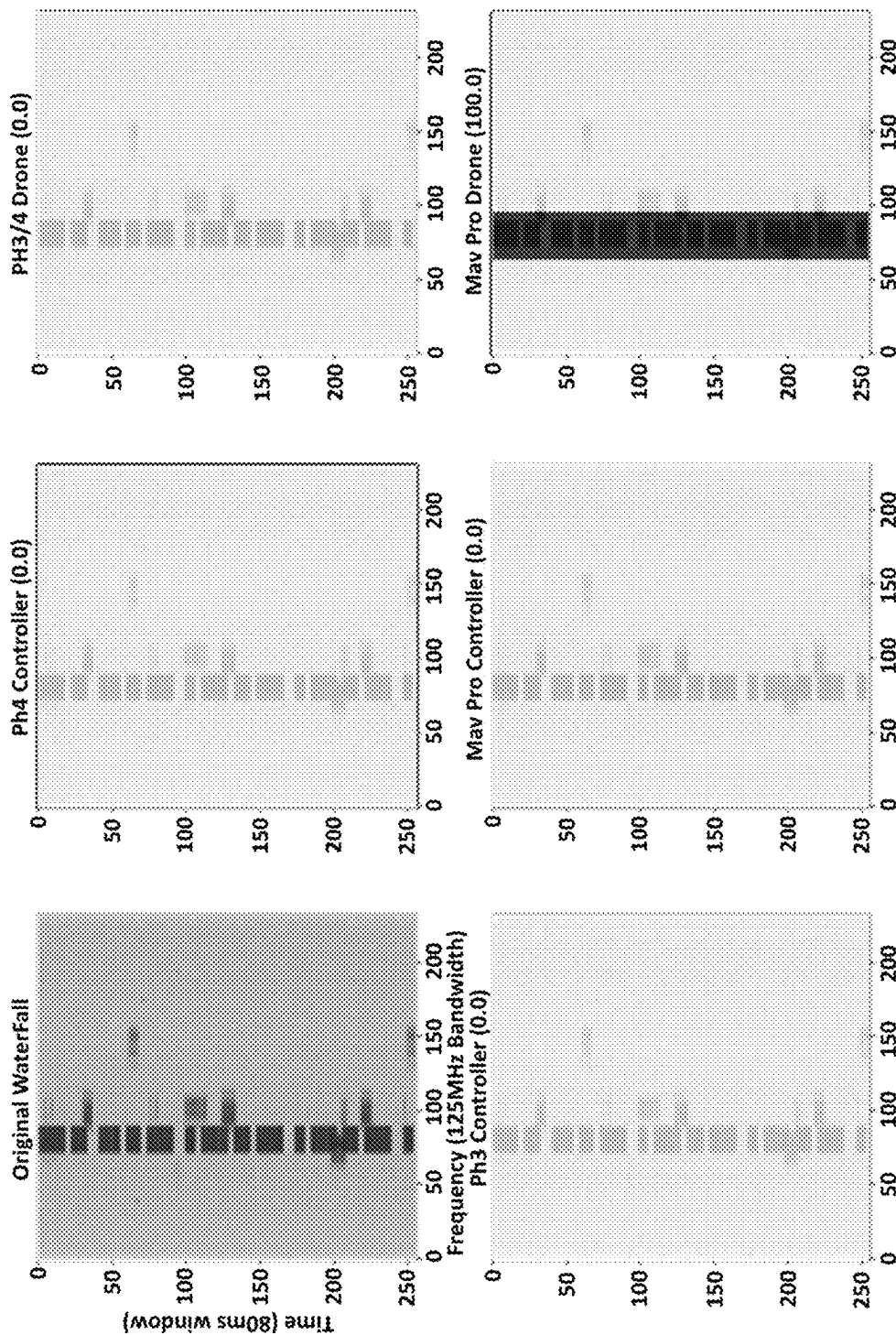


FIG. 12

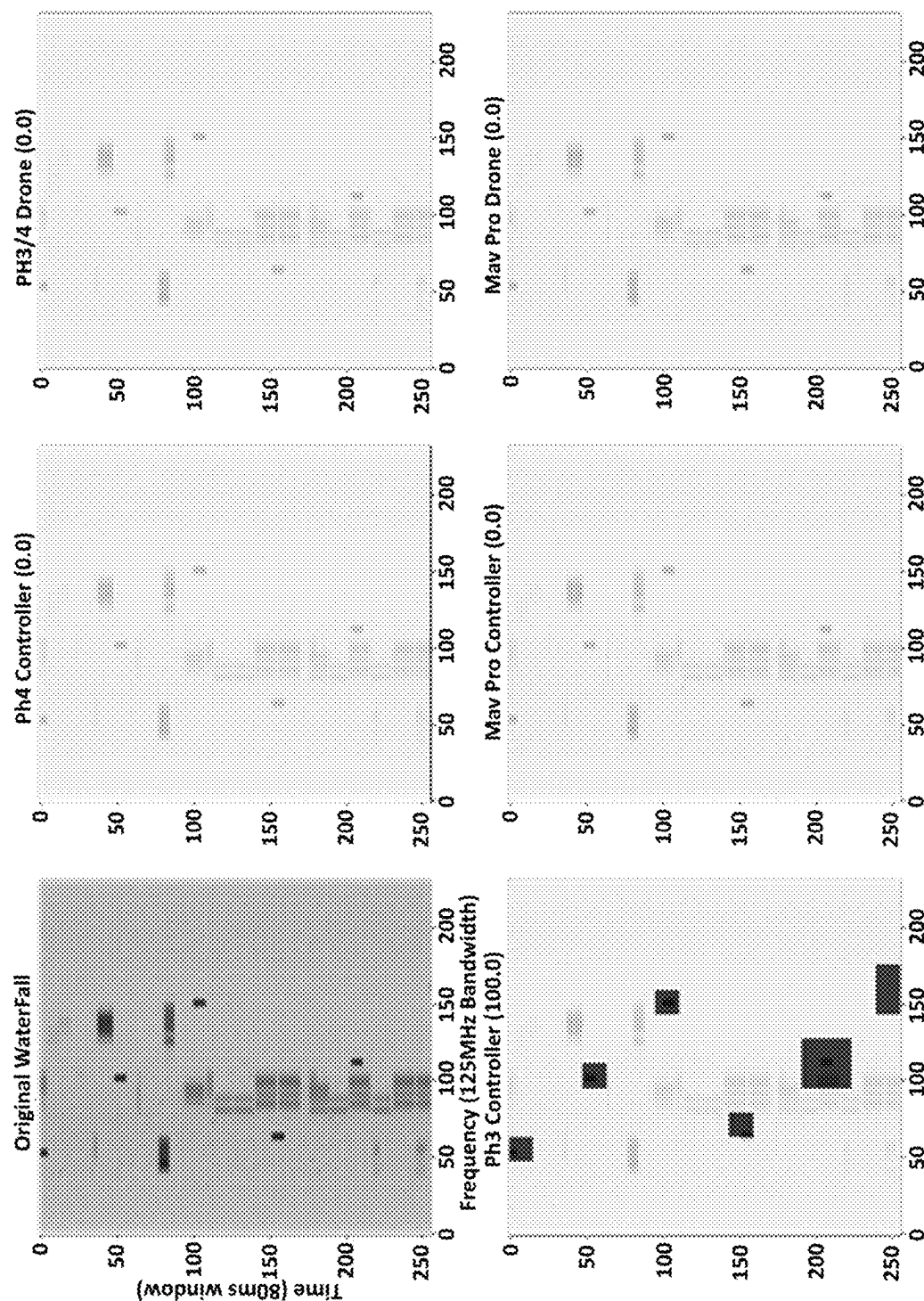


FIG. 13

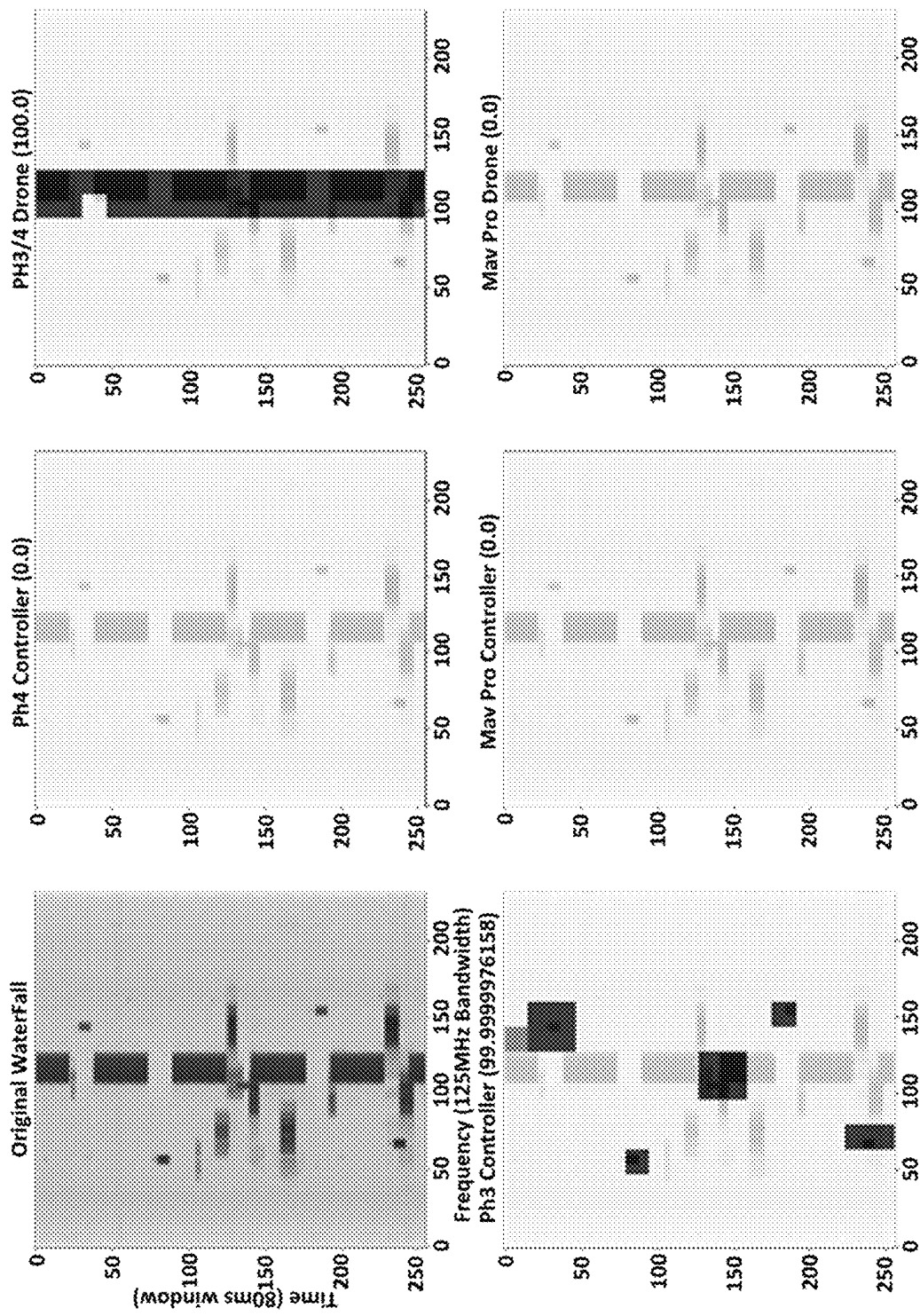


FIG. 14

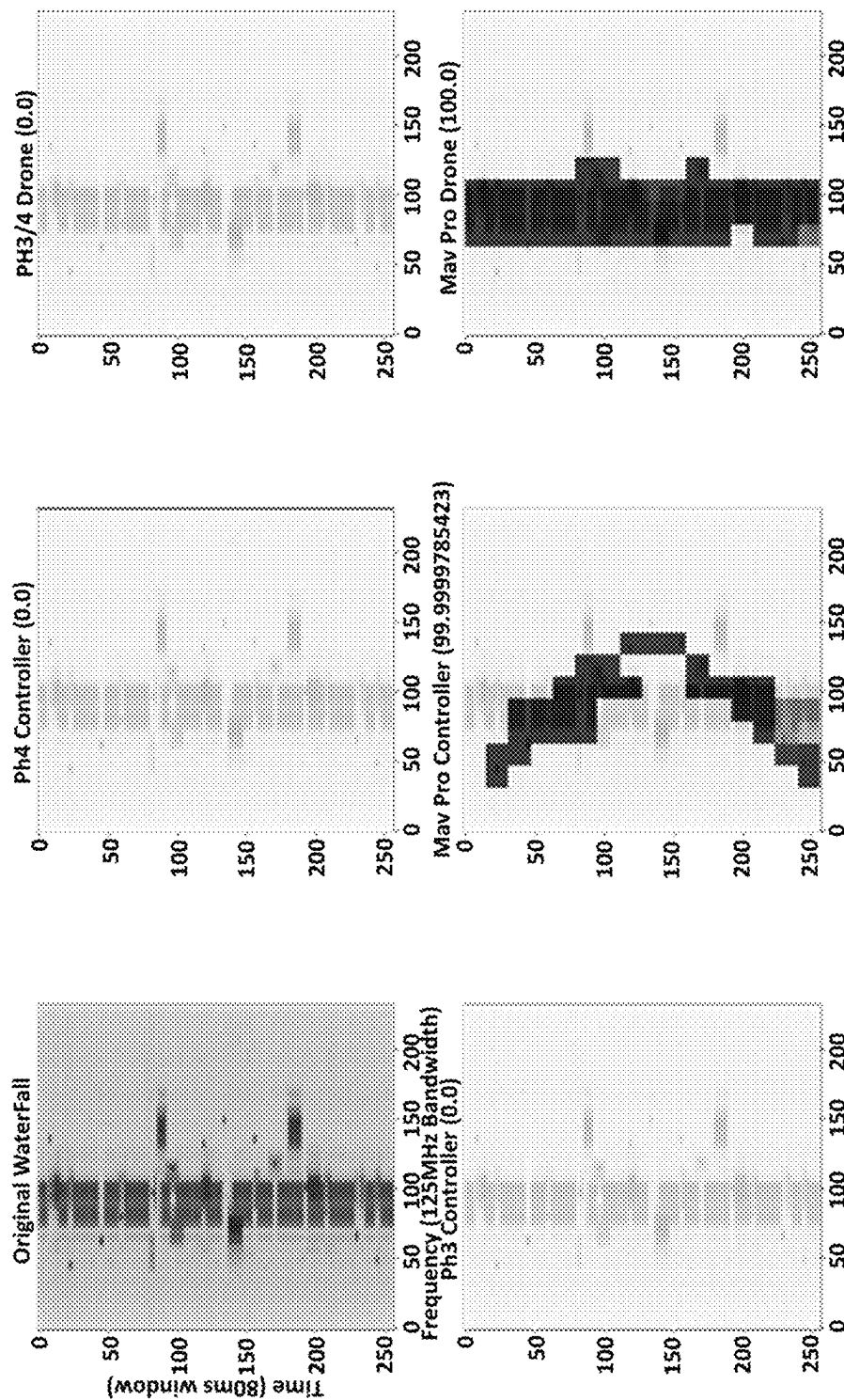


FIG. 15

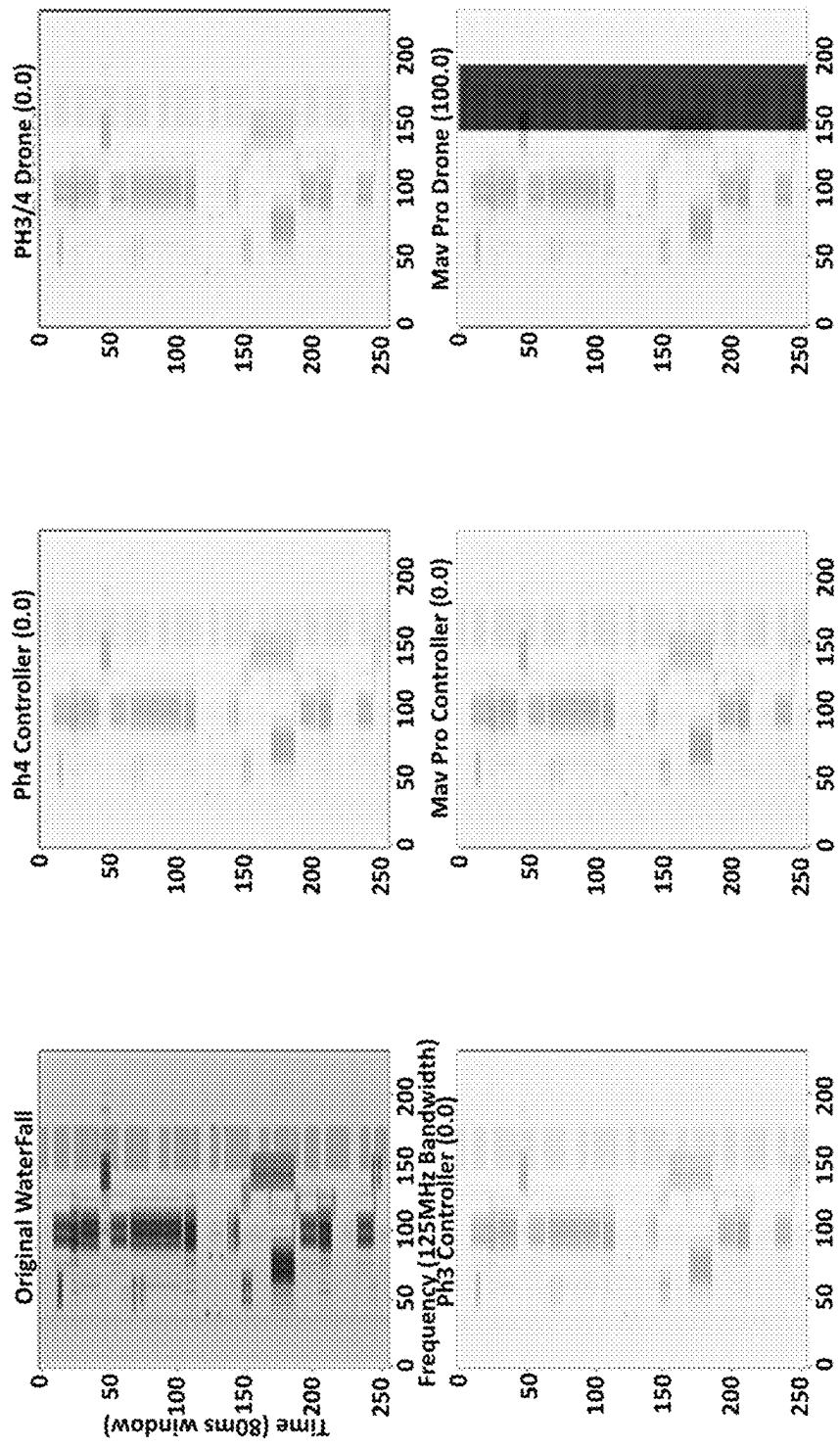


FIG. 16

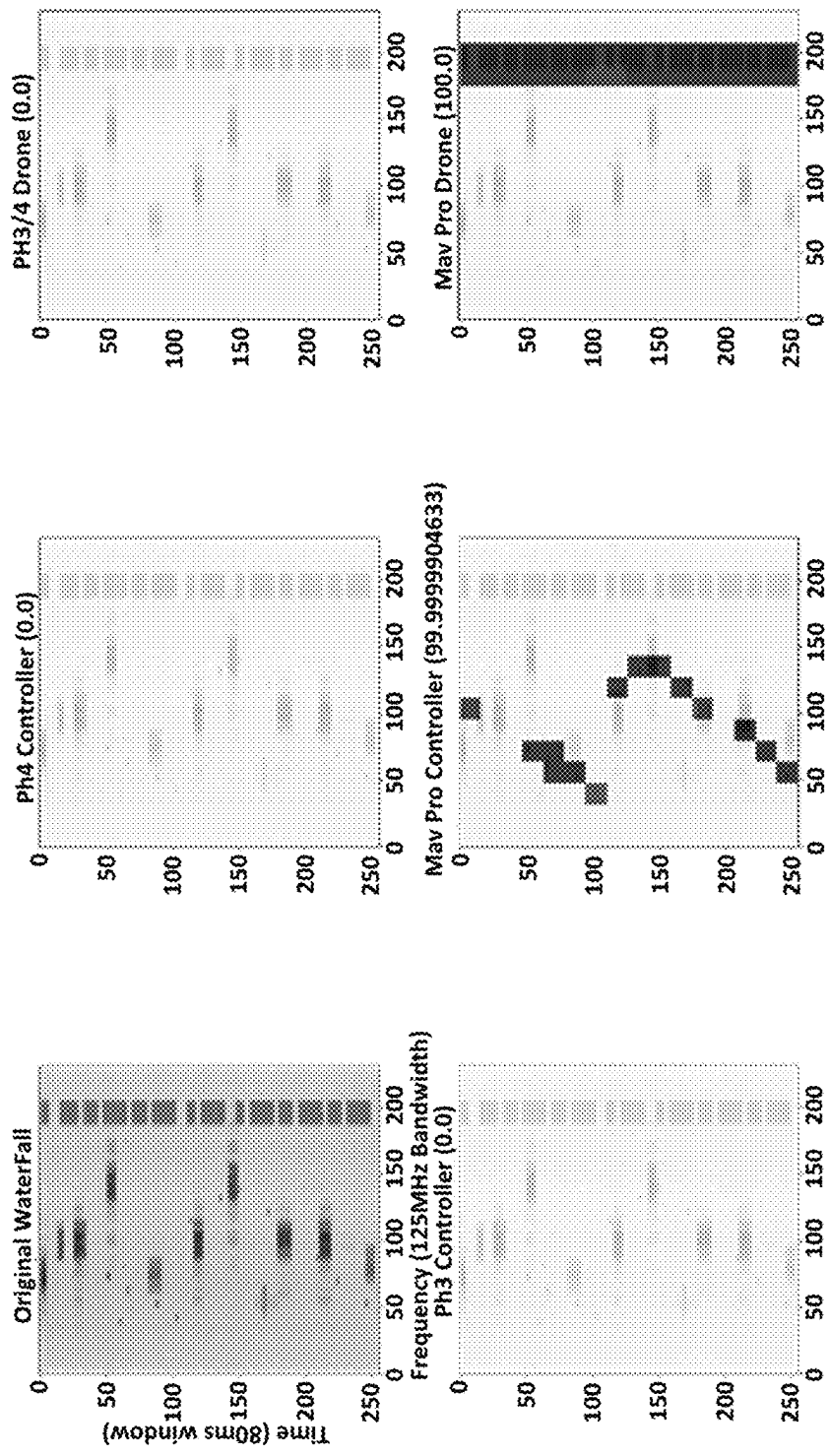


FIG. 17

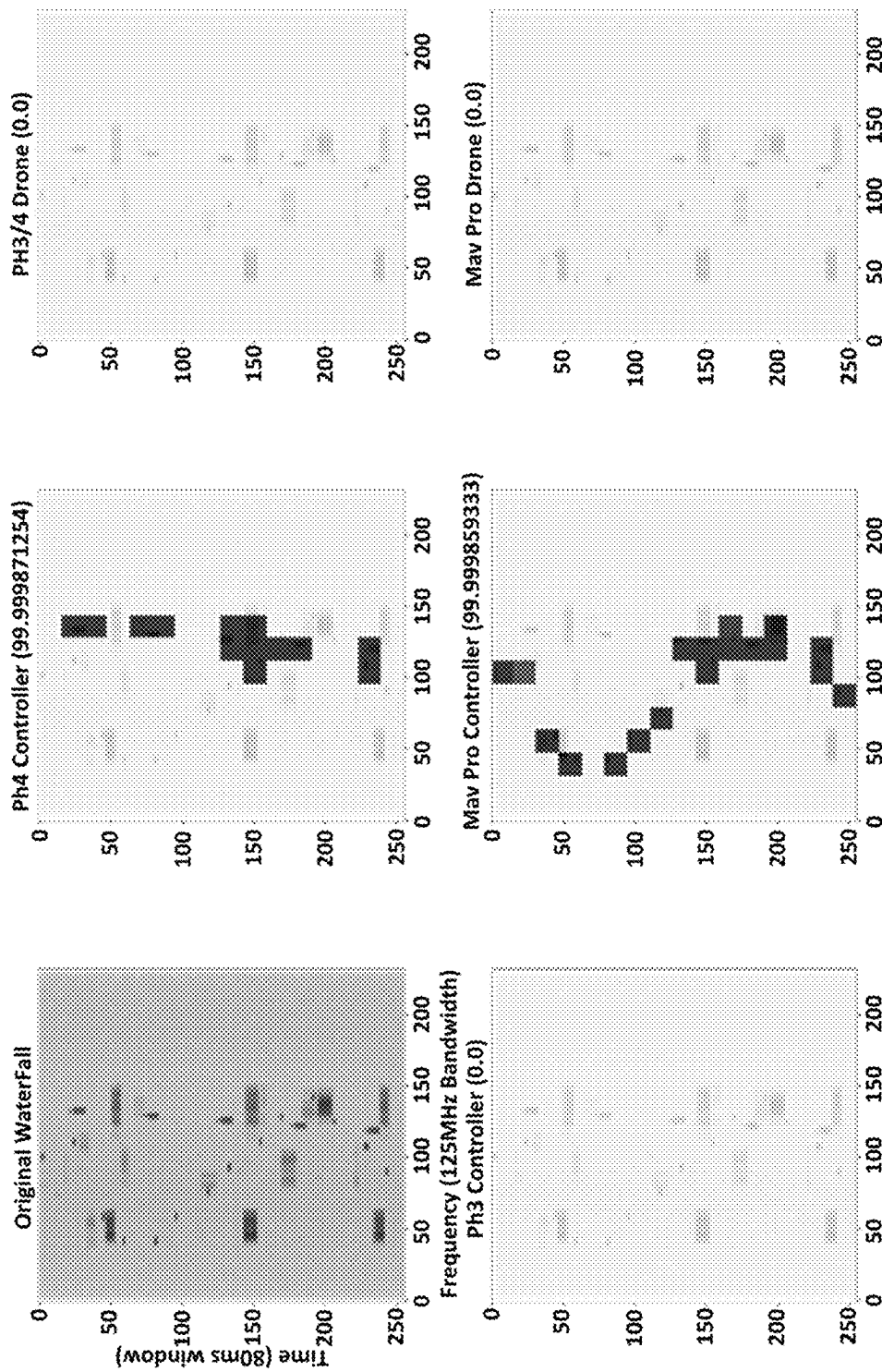


FIG. 18

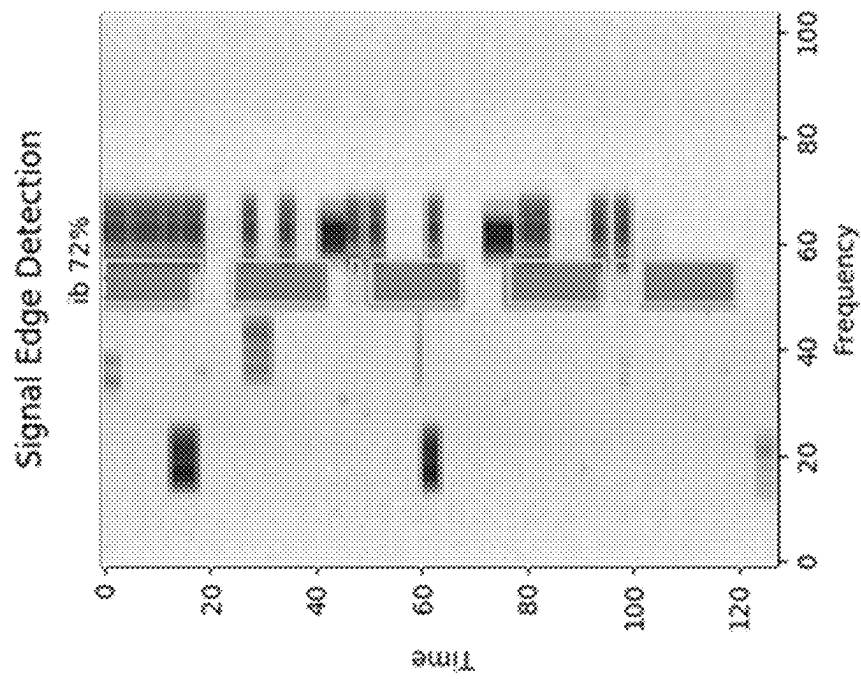


FIG. 19

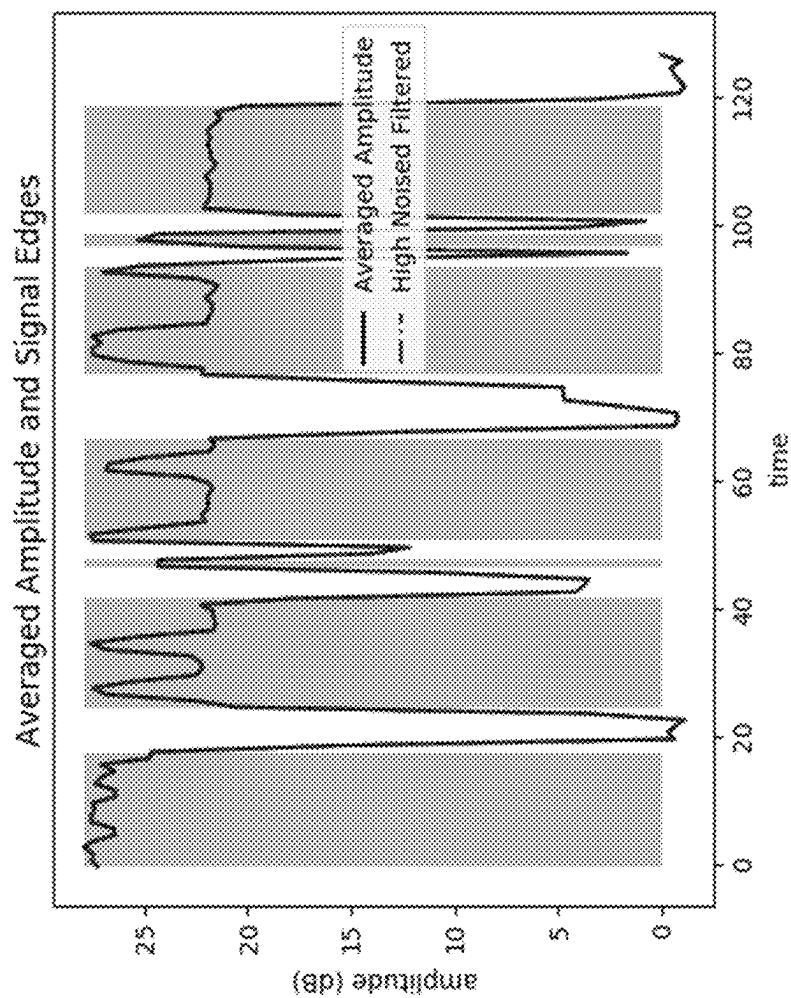


FIG. 20

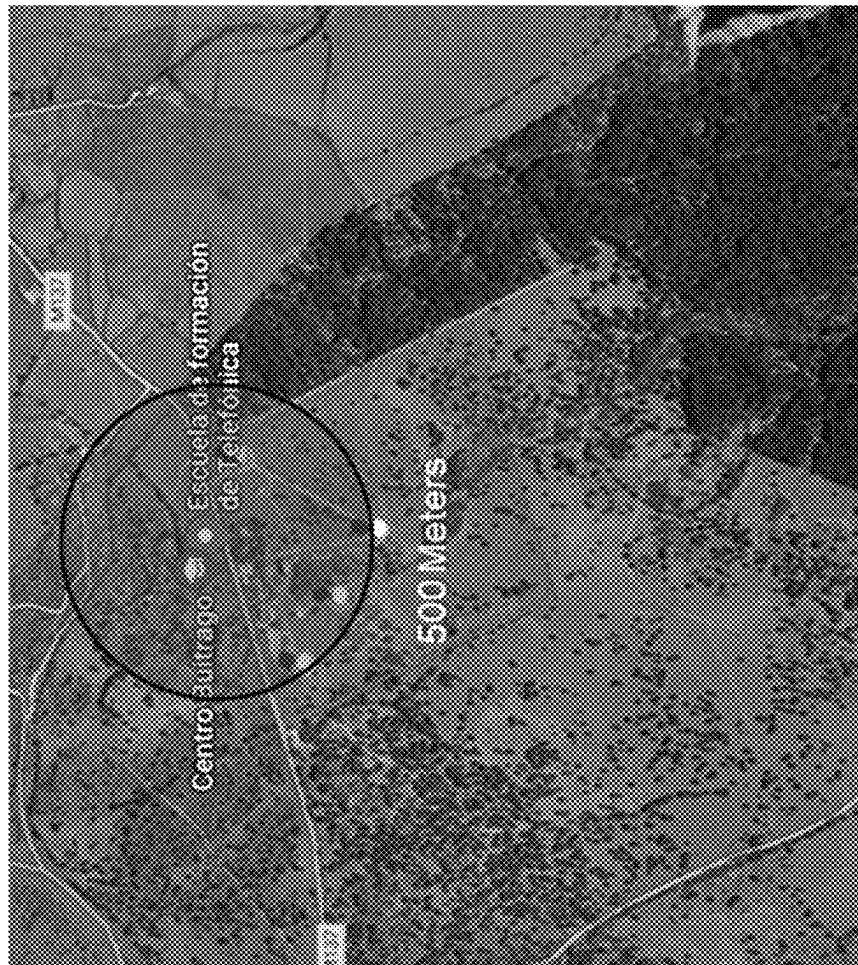


FIG. 21

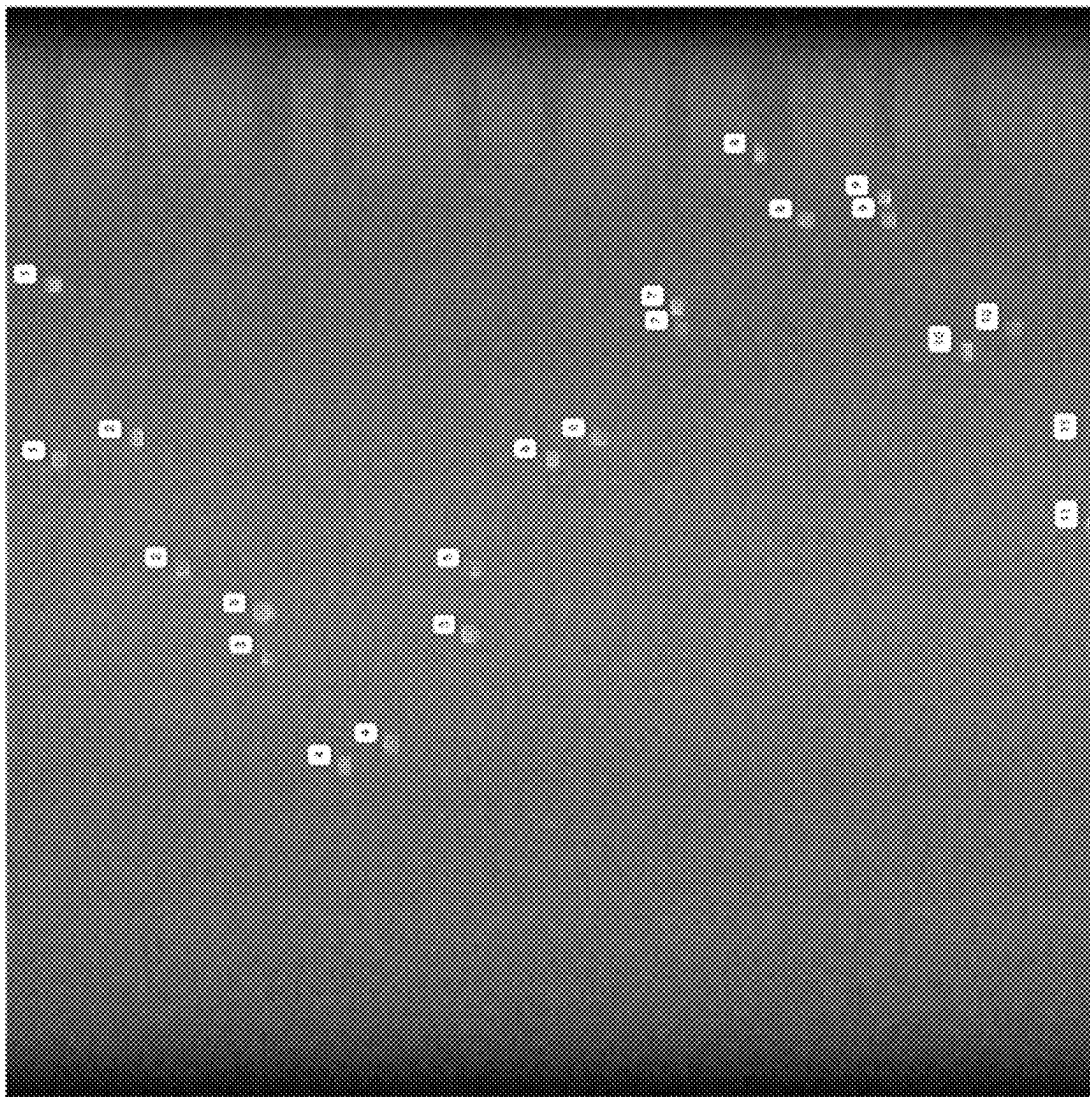


FIG. 22

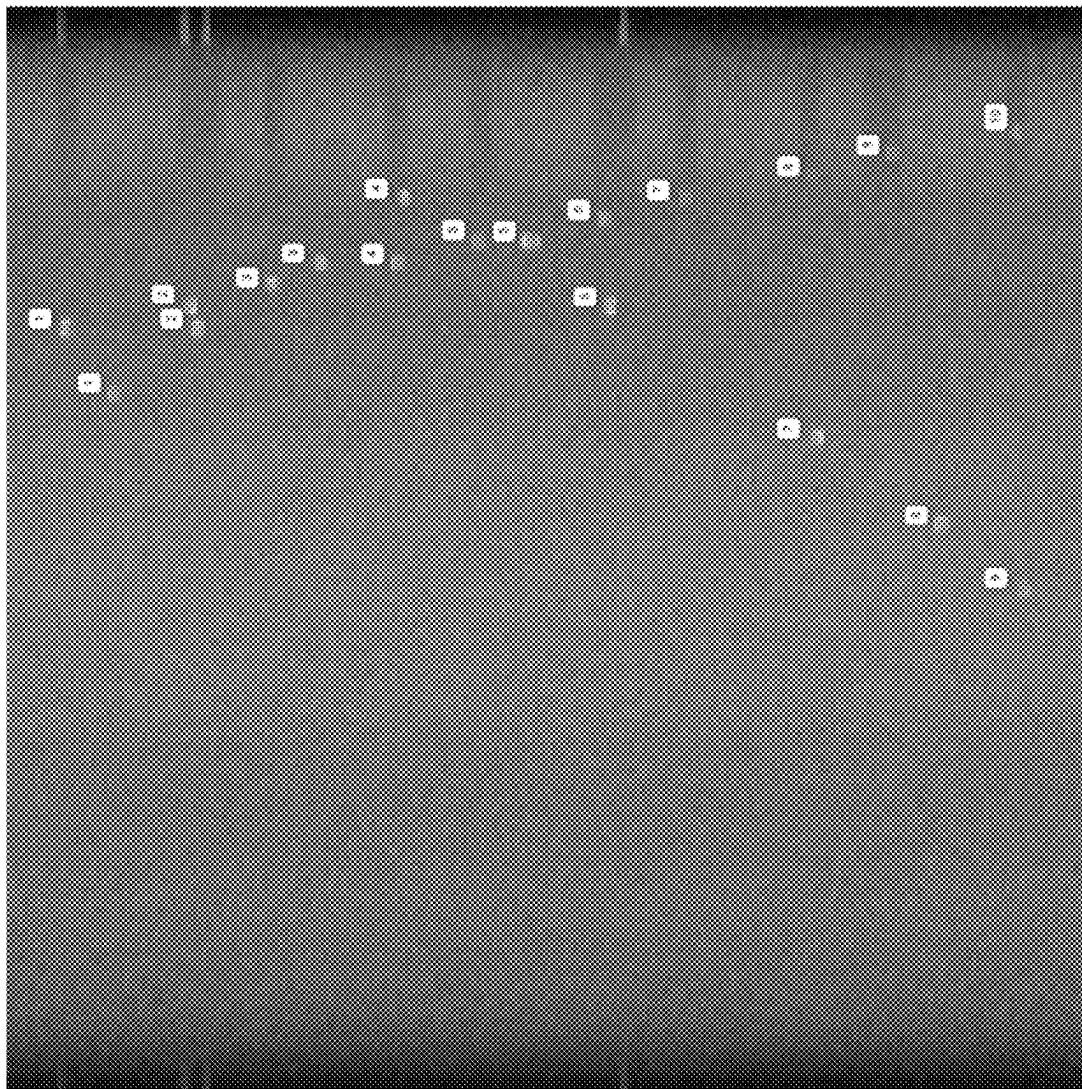


FIG. 23

1

UNMANNED VEHICLE RECOGNITION AND THREAT MANAGEMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application relates to and claims priority from the following applications. This application is a continuation of U.S. patent application Ser. No. 18/985,777, filed Dec. 18, 2024, which is a continuation of U.S. patent application Ser. No. 18/789,011, filed Jul. 30, 2024, which is a continuation of U.S. patent application Ser. No. 18/785,936, filed Jul. 26, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/775,710, filed Jul. 17, 2024, which is a continuation-in-part of U.S. patent application Ser. No. 18/428,606, filed Jan. 31, 2024, which is a continuation of U.S. patent application Ser. No. 18/374,376, filed Sep. 28, 2023, which is a continuation of U.S. patent application Ser. No. 18/142,904, filed May 3, 2023, which is a continuation of U.S. patent application Ser. No. 17/991,348, filed Nov. 21, 2022, which is a continuation of U.S. patent application Ser. No. 17/735,615, filed May 3, 2022, which is a continuation of U.S. patent application Ser. No. 17/190,048 filed Mar. 2, 2021, which is a continuation of U.S. patent application Ser. No. 16/732,811 filed Jan. 2, 2020, which is a continuation of U.S. patent application Ser. No. 16/275,575 filed Feb. 14, 2019, which claims the benefit of U.S. Provisional Application 62/632,276 filed Feb. 19, 2018. U.S. patent application Ser. No. 16/275,575 also claims priority from and is a continuation-in-part of U.S. patent application Ser. No. 16/274,933 filed Feb. 13, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 16/180,690 filed Nov. 5, 2018, which is a continuation-in-part of U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017. U.S. patent application Ser. No. 16/180,690 also claims priority from U.S. Provisional Patent Application No. 62/722,420 filed Aug. 24, 2018. U.S. patent application Ser. No. 16/274,933 also claims the benefit of U.S. Provisional Application 62/632,276 filed Feb. 19, 2018. Each of the above-mentioned applications is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to systems and methods for unmanned vehicle recognition and threat management. More particularly, the systems and methods of the present invention are directed to unmanned vehicle detection, classification and direction finding.

2. Description of the Prior Art

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have become readily available in commercial and retail stores. Detailed schematics for their control systems are available from many manufacturers and the internet along with Software Development Kits (SDKs). Rapid modifications are enabled by advancements in various technologies (e.g., 3D printing). UAVs can be modified to deploy dangerous actions and threaten societal securities. For example, UAVs can be modified to deliver dangerous payloads. It is no longer a question of if, it is now a question of when. Thus, it is imperative that organizations and govern-

2

ments take steps to protect critical assets (e.g., ports, power plants), structures (e.g., buildings, stadiums), and personnel and their citizens.

Exemplary U.S. Patent Documents relevant to the prior art include:

U.S. Pat. No. 9,862,489 for "Method and apparatus for drone detection and disablement" by inventors Lee Weinstein et al., filed Feb. 7, 2016 and issued Jan. 9, 2018, describes a method and apparatus for detection and disablement of an unidentified aerial vehicle (UAV) includes arrays of antenna elements receiving in two modalities (for instance radio frequency (RF) and acoustic modalities, or RF and optical modalities). Signal processing of outputs from multiple antenna arrays locates a potential UAV at specific coordinates within a volume of space under surveillance, and automatically aims video surveillance and a short-range projectile launcher at the UAV, and may automatically fire the projectile launcher to down the UAV.

U.S. Pat. No. 9,858,947 for "Drone detection and classification methods and apparatus" by inventors Brian Hearing et al., filed Nov. 24, 2015 and issued Jan. 2, 2018, describes a system, method, and apparatus for drone detection and classification. An example method includes receiving a sound signal in a microphone and recording, via a sound card, a digital sound sample of the sound signal, the digital sound sample having a predetermined duration. The method also includes processing, via a processor, the digital sound sample into a feature frequency spectrum. The method further includes applying, via the processor, broad spectrum matching to compare the feature frequency spectrum to at least one drone sound signature stored in a database, the at least one drone sound signature corresponding to a flight characteristic of a drone model. The method moreover includes, conditioned on matching the feature frequency spectrum to one of the drone sound signatures, transmitting, via the processor, an alert.

U.S. Pat. No. 9,767,699 for "System for and method of detecting drones" by inventors John W. Borghese et al., filed May 14, 2015 and issued Sep. 19, 2017, describes an apparatus and method can provide a warning of a drone or unmanned aerial vehicle in the vicinity of an airport. The apparatus can include at least one antenna directionally disposed at an along the approach or departure path and a detector configured to provide a warning of a presence of sense an unmanned aerial or drone. The warning can be provided in response to a radio frequency signal received by the at least one of the antenna being in a frequency band associated with a transmission frequency for the unmanned aerial vehicle or drone or in a frequency band associated with interaction from receive circuitry of the unmanned aerial vehicle or drone.

U.S. Pat. No. 9,715,009 for "Deterrent for unmanned aerial systems" by inventors Dwaine A. Parker et al., filed Dec. 2, 2016 and issued Jul. 25, 2017, describes a system for providing an integrated multi-sensor detection and countermeasure against commercial unmanned aerial systems/vehicles and includes a detecting element, a tracking element, an identification element, and an interdiction element. The detecting element detects an unmanned aerial vehicle in flight in the region of, or approaching, a property, place, event or very important person. The tracking element determines the exact location of the unmanned aerial vehicle. The identification/classification element utilizing data from the other elements generates the identification and threat assessment of the UAS. The interdiction element, based on automated algorithms can either direct the unmanned aerial vehicle away from the property, place, event or very impor-

tant person in a non-destructive manner, or can disable the unmanned aerial vehicle in a destructive manner. The interdiction process may be over ridden by intervention by a System Operator/HiL.

U.S. Pat. No. 9,529,360 for “System and method for detecting and defeating a drone” by inventors Howard Melamed et al., filed Apr. 22, 2015 and issued Dec. 27, 2016, describes a system for detecting and defeating a drone. The system has a detection antenna array structured and configured to detect the drone and the drone control signal over a 360 degree field relative to the detection antenna array including detecting the directionality of the drone. The system also includes a neutralization system structured and configured in a communicating relation with the detection antenna array. The neutralization system has a transmission antenna structured to transmit an override signal aimed at the direction of the drone, an amplifier configured to boost the gain of the override signal to exceed the signal strength of the drone control signal, and a processing device configured to create and effect the transmission of the override signal. The patent also discloses a method for detecting and defeating a drone.

U.S. Publication No. 2017/0358103 for “Systems and Methods for Tracking Moving Objects” by inventors Michael Shao et al., filed Jun. 9, 2017 and published Dec. 14, 2017, describes systems and methods for tracking moving objects. The publication discloses an object tracking system comprises a processor, a communications interface, and a memory configured to store an object tracking application. The object tracking application configures the processor to receive a sequence of images; estimate and subtract background pixel values from pixels in a sequence of images; compute sets of summed intensity values for different per frame pixel offsets from a sequence of images; identify summed intensity values from a set of summed intensity values exceeding a threshold; cluster identified summed intensity values exceeding the threshold corresponding to single moving objects; and identify a location of at least one moving object in an image based on at least one summed intensity value cluster.

U.S. Publication No. 2017/0261613 for “Counter drone system” by inventor Brian R. Van Voorst, filed Feb. 27, 2017 and published Sep. 14, 2017, describes a counter drone system that includes a cueing sensor to detect the presence of an object wherein the cueing sensor cues the presence of a target drone, a long range LIDAR system having a sensor pointed in a direction of the target drone to acquire and track at long range the target drone to provide an accurate location of the target drone wherein once a track is acquired, the motion of the target drone is used to maintain the track of the target drone and a threat detector wherein LIDAR data is provided to the threat detector to determine if the target drone is a threat.

U.S. Publication No. 2017/0261604 for “Intercept drone tasked to location of lidar tracked drone” by inventor Brian Van Voorst, filed Feb. 27, 2017 and published Sep. 14, 2017, describes a system that includes a long range LIDAR tracking system to track a target drone and provide detection and tracking information of the target drone; a control system to process the detection and tracking information and provide guidance information to intercept the target drone; and a high powered intercept drone controlled by supervised autonomy, the supervised autonomy provided by processing the detection and tracking information of the target drone and sending guidance information to the intercept drone to direct the intercept drone to the target drone.

U.S. Publication No. 2017/0039413 for “Commercial drone detection” by inventor Gary J. Nadler, filed Aug. 3, 2015 and published Feb. 9, 2017, describes a method of capturing the presence of a drone, including: collecting, using at least one sensor, data associated with an aerial object; analyzing, using a processor, the data to determine at least one characteristic of the aerial object; accessing, in a database, a library of stored characteristics of commercially available drones; determining, based on the analyzing, if the at least one characteristic of the aerial object matches a characteristic of a commercially available drone; and responsive to the determining, generating an indication of a positive match.

U.S. Pat. No. 11,663,992 for Systems and Methods for Detecting, Monitoring, and Mitigating the Presence of Unauthorized Drones by inventors Jordan, et al., Jun. 29, 2020 and published May 23, 2023, is directed to Systems and methods for detecting, monitoring, and mitigating the presence of a drone are provided herein. In one aspect, a system for detecting presence of a drone includes a radio-frequency (RF) receiver. The system can further include a processor and a computer-readable memory in communication with the processor and having stored thereon computer-executable instructions to cause the at least one processor to receive a set of samples from the RF receiver for a time interval, obtain predetermined data of expected communication protocols used between the drone and a controller, and determine whether the RF signal corresponds to one of the expected communication protocols by comparing the samples of the RF signal to the predetermined data and decoding the RF signal. In further aspects the system extracts a unique identifier of the drone based at least partially on the decoded RF signal.

U.S. Pat. No. 11,190,233 for Systems and Methods for Detecting, Monitoring, and Mitigating the Presence of a Drone Using Frequency Hopping by inventors Fang-Hsuan Lo, et al., filed May 28, 2020 and published Nov. 10, 2021, is directed to systems and methods for detecting, monitoring, and mitigating the presence of a drone are provided herein. In one aspect, a system for detecting presence of a one or more drones includes a radio-frequency (RF) receiver configured to receive an RF signal transmitted between a drone and a controller. The system can further include a processor and a computer-readable memory in communication with the processor and having stored thereon computer-executable instructions to cause the at least one processor to receive a set of samples from the RF receiver for a time interval, the set of samples comprising samples of the first RF signal, obtain a parameter model of the first frequency hopping parameters, and fit the parameter model to the set of samples.

U.S. Pat. No. 10,698,076 for Radio Frequency Signal Transmission Detector and Locator by inventors Jones, et al., filed Aug. 2, 2018 and published Jun. 10, 2020, is directed to The present invention is a system and method for detecting and locating the transmission of radio frequency signals from within a defined geographical area. The system uses statistical confidence limits to detect outliers caused by transmissions in the defined geographical area. The source of the transmission can then be located with triangulation.

U.S. Pat. No. 10,540,905 for Systems, Aircrafts, and Methods for Drone Detection and Collision Avoidance by inventors Bohanan, et al., filed Mar. 28, 2018 and published Jan. 1, 2020, is directed to a system and a method for drone detection and collision avoidance, particularly for use in an aircraft, is provided. The system includes, but is not limited to a sensor, a processor, and an avoidance unit comprising a

control unit. The sensor is configured to detect a drone signal in a predetermined space and to transmit the drone signal to the processor. The processor is configured to determine the presence of a drone in the predetermined space based on the drone signal. The processor is configured to transmit a command to the avoidance unit when the processor determines the presence of a drone. The control unit is configured to receive the command and to generate a warning signal in response to receiving the command.

U.S. Pat. No. 9,529,360 for System and Method for Detecting and Defeating a Drone by inventors Melamed, et al., filed Apr. 22, 2015 and published Dec. 7, 2016, is directed to The invention is directed to a system for detecting and defeating a drone. The system has a detection antenna array structured and configured to detect the drone and the drone control signal over a 360 degree field relative to the detection antenna array including detecting the directionality of the drone. The system also includes a neutralization system structured and configured in a communicating relation with the detection antenna array. The neutralization system has a transmission antenna structured to transmit an override signal aimed at the direction of the drone, an amplifier configured to boost the gain of the override signal to exceed the signal strength of the drone control signal, and a processing device configured to create and effect the transmission of the override signal. The invention is also directed to a method for detecting and defeating a drone.

SUMMARY OF THE INVENTION

The present invention provides systems and methods for unmanned vehicle recognition. In one embodiment, a multiplicity of receivers captures RF data and transmits the RF data to at least one node device. The at least one node device comprises a signal processing engine, a detection engine, a classification engine, and a direction finding engine. The at least one node device is configured with an artificial intelligence algorithm. The detection engine and classification engine are trained to detect and classify signals from unmanned vehicles and their controllers based on processed data from the signal processing engine. The direction finding engine is operable to provide lines of bearing for detected unmanned vehicles. A display and control unit is in network communication with the at least one node device for displaying locations and other related data for the detected unmanned vehicles.

In one embodiment, the present invention includes a system for signal identification in a radiofrequency (RF) environment, including at least one node device including a processor and at least one memory in communication with at least one RF receiver, wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device, wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is visually represented as at least one waterfall image, wherein the at least one node device is operable to analyze the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

In another embodiment, the present invention includes an apparatus for signal identification in a radiofrequency (RF) environment, including a node device including a processor and at least one memory, wherein the node device is operable to receive RF data from at least one RF receiver, wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, wherein the node device waterfall image is analyzed using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

In yet another embodiment, the present invention includes a method for signal identification in a radiofrequency (RF) environment, including at least one RF receiver capturing Fast Fourier Transform (FFT) data in the RF environment and transmitting the FFT data to at least one node device, the at least one node device averaging the FFT data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, the at least one node device analyzing the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise, and the at least one node device creating at least one analyzed waterfall image based on the identification of the at least one signal, the at least one signal type, and/or the noise, wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system for unmanned vehicle recognition according to one embodiment of the present invention.

FIG. 2 illustrates signal characterization within a spectrum from 700 MHz to 900 MHz.

FIG. 3 is an illustration of Phantom 4 controller and drone signals.

FIG. 4 is a human interface display for drone detection according to one embodiment of the present invention.

FIG. 5 shows a setup for RF data capture in an Anechoic Chamber according to one embodiment of the present invention.

FIG. 6 illustrates a simulation for fading and channel hopping of an OcuSync drone.

FIG. 7 is an illustration of an inception based convolutional neural network.

FIG. 8 illustrates a scenario for an RF environment with just noise.

FIG. 9 illustrates a scenario for an RF environment with a Phantom 4 controller and drone.

FIG. 10 illustrates a scenario for an RF environment with two Mavic Pro drones.

FIG. 11 illustrates a scenario for an RF environment with a Mavic Pro controller only.

FIG. 12 illustrates a scenario for an RF environment with one Mavic Pro drone only.

FIG. 13 illustrates a scenario for an RF environment with one Phantom 3 controller only.

FIG. 14 illustrates a scenario for an RF environment with a Phantom 3 controller and drone.

FIG. 15 illustrates a scenario for an RF environment with a Mavic Pro controller and drone in wideband mode.

FIG. 16 illustrates a scenario for an RF environment with a Mavic Pro drone in wideband mode.

FIG. 17 illustrates a scenario for an RF environment with a Mavic Pro controller and drone.

FIG. 18 illustrates a scenario for an RF environment with a Mavic Pro controller and a Phantom 4 controller.

FIG. 19 is an illustration of identifying drones and controllers based on signal edge detection.

FIG. 20 is an illustration with averaged signal amplitudes and signal edges according to one embodiment of the present invention.

FIG. 21 displays a detection range of less than 500 meters based on equipment specification and location.

FIG. 22 displays a training data set containing two separate drone devices operating on the same type of radio protocol according to one embodiment of the present invention.

FIG. 23 displays a training data set containing two separate drone devices operating on different types of radio protocols according to one embodiment of the present invention.

DETAILED DESCRIPTION

The present invention provides systems and methods for unmanned vehicle recognition. The present invention relates to automatic signal detection, temporal feature extraction, geolocation, and edge processing disclosed in U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017, U.S. patent application Ser. No. 15/478,916 filed Apr. 4, 2017, U.S. patent application Ser. No. 15/681,521 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,540 filed Aug. 21, 2017, and U.S. patent application Ser. No. 15/681,558 filed Aug. 21, 2017, each of which is incorporated herein by reference in its entirety.

In one embodiment, the present invention includes a system for signal identification in a radiofrequency (RF) environment, including at least one node device including a processor and at least one memory in communication with at least one RF receiver, wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device, wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile, wherein the at least one tile is visually represented as at least one waterfall image, wherein the at least one node device is operable to analyze the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

In another embodiment, the present invention includes an apparatus for signal identification in a radiofrequency (RF) environment, including a node device including a processor and at least one memory, wherein the node device is operable to receive RF data from at least one RF receiver, wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least

one tile, wherein the at least one tile is represented as at least one waterfall image, wherein the node device waterfall image is analyzed using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise to create at least one analyzed waterfall image, and wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

In yet another embodiment, the present invention includes a method for signal identification in a radiofrequency (RF) environment, including at least one RF receiver capturing Fast Fourier Transform (FFT) data in the RF environment and transmitting the FFT data to at least one node device, the at least one node device averaging the FFT data derived from the RF data into at least one tile, wherein the at least one tile is represented as at least one waterfall image, the at least one node device analyzing the at least one waterfall image using machine learning (ML) or at least one convolutional neural network (CNN) to identify at least one signal, at least one signal type, and/or noise, and the at least one node device creating at least one analyzed waterfall image based on the identification of the at least one signal, the at least one signal type, and/or the noise, wherein the at least one analyzed waterfall image includes a visual indication of the at least one signal, the at least one signal type, and/or the noise.

Currently, commercial and retail UAVs dominate frequencies including 433 MHz industrial, scientific, and medical radio band (ISM Band) Region 1, 900 MHz ISM Band Region 1,2,3 (varies by country), 2.4 GHz (channels 1-14), 5 GHz (channels 7-165 most predominant), and 3.6 GHz (channels 131-183). Modulation types used by commercial and retail UAVs include Direct Sequence Spread Spectrum (DSSS), Orthogonal Frequency Division Multiplexing (OFDM), Frequency Hopping Spread Spectrum (FHSS), Fataba Advanced Spread Spectrum Technology (FASST).

Many counter UAV systems in the prior art focus on the 2.4 GHz and 5.8 GHz bands utilizing demodulation and decryption of radio frequency (RF) signals to detect and analyze each signal to determine if it is related to a UAV.

The present invention provides systems and methods for unmanned vehicle recognition including detection, classification and direction finding. Unmanned vehicles comprise aerial, terrestrial or water borne unmanned vehicles. The systems and methods for unmanned vehicle recognition are operable to counter threats from the aerial, terrestrial or water borne unmanned vehicles.

In one embodiment, a multiplicity of receivers captures RF data and transmits the RF data to at least one node device. The at least one node device comprises a signal processing engine, a detection engine, a classification engine, and a direction finding engine. The at least one node device is configured with an artificial intelligence algorithm. The detection engine and classification engine are trained to detect and classify signals from unmanned vehicles and their controllers based on processed data from the signal processing engine. The direction finding engine is operable to provide lines of bearing for detected unmanned vehicles. A display and control unit is in network communication with the at least one node device for displaying locations and other related data for the detected unmanned vehicles.

In one embodiment, the present invention provides systems and methods for unmanned vehicle (UV) recognition in a radio frequency (RF) environment. A multiplicity of RF receivers and a displaying device are in network communication with a multiplicity of node devices. The multiplicity

of RF receivers is operable to capture the RF data in the RF environment, convert the RF data to fast Fourier transform (FFT) data, and transmit the FFT data to the multiplicity of node devices. The multiplicity of node devices each comprises a signal processing engine, a detection engine, a classification engine, a direction-finding engine, and at least one artificial intelligence (AI) engine. The signal processing engine is operable to average the FFT data into at least one tile. The detection engine is operable to group the FFT data into discrete FFT bins over time, calculate average and standard deviation of power for the discrete FFT bins, and identify at least one signal related to at least one UV and/or corresponding at least one UV controller. The at least one AI engine is operable to generate an output for each of the at least one tile to identify at least one UV and corresponding at least one UV controller with a probability, and calculate an average probability based on the output from each of the at least one tile. The classification engine is operable to classify the at least one UV and/or the at least one UV controller by comparing the at least one signal to classification data stored in a classification library in real time or near real time. The direction-finding engine is operable to calculate a line of bearing for the at least one UV. The displaying device is operable to display a classification of the at least one UV and/or the at least one UV controller and/or the line of bearing of the at least one UV. Each of the at least one tile is visually represented in a waterfall image via a graphical user interface on the displaying device.

FIG. 1 illustrates a system for unmanned vehicle recognition according to one embodiment of the present invention. The system includes a multiplicity of antennas, a receiver and processing unit, and a display and control unit. In one embodiment, there are four multiband omnidirectional antennas. In one embodiment, three multiband omnidirectional antennas are positioned to form an equilateral with 6 meters spacing as illustrated in FIG. 1 as an example. The receiver and processing unit includes a signal processing engine, a UAV detection engine, a UAV classification engine, a direction finding processing engine, and an internal Global Positioning System (GPS). The receiver and processing unit is operable to receive RF data from the antennas and automatically process the RF data for UAV detection and classification and direction finding. The display and control unit includes a human interface display. In one embodiment, the human interface display is provided by a remote web-based interface. The display and control unit is operable to display lines of bearings for detected UAVs and controllers, classification for detected UAVs and controllers, received signal strength (RSS) values, and operating frequencies. In one embodiment, the display and control unit is SigBase 4000 as shown in FIG. 1. In another embodiment, any computer, laptop, or tablet configured with the human interface display of the present invention is operable to function as a display and control unit. In one embodiment, the receiver and processing unit is a node device, and there are multiple node devices communicating with each other and forming a group of networked nodes for UAV detection, classification, and direction finding.

The present invention provides a more efficient methodology for UAV detection and identification, which takes advantage of Fast Fourier Transform (FFT) over a short period of time and its derivation. RF data received from antennas are directly converted to FFT data with finer granularity. This allows rapid identification of protocols used by high threat drones without demodulation, and the identification is probability based. An analytics engine is operable to perform near real-time analysis and characterize

signals within the spectrum under observation. FIG. 2 illustrates signal characterization within a spectrum from 700 MHz to 900 MHz. Temporal feature extraction is applied for signal characterization, which is described in U.S. patent application Ser. No. 15/412,982 filed Jan. 23, 2017, U.S. patent application Ser. No. 15/681,521 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,540 filed Aug. 21, 2017, U.S. patent application Ser. No. 15/681,558 filed Aug. 21, 2017, each of which is incorporated herein by reference in its entirety.

Advantageously, multiple receivers in the present invention work together to ingest spectral activities across large blocks of spectrum. The multiple receivers have an instantaneous bandwidth from 40 MHz to 500 MHz. In one embodiment, the multiple receivers are configurable in 40 MHz and 125 MHz segment building blocks. Input data are converted directly to FFT data and fed into process engines, which significantly decreases latency. The process engines are designed for rapid identification of signals of interest (SOI). When an SOI is detected, a direction finding process is initiated autonomously. In one embodiment, the direction finding process is configurable by an operator.

There are multiple types of communications links utilized for command and control of an unmanned vehicle. Although several cost-effective radio communication (RC) protocols are gaining global popularity, WI-FI is still the most popular protocol for command and control of UAVs and camera systems. A remote controller of a UAV acts as a WI-FI access point and the UAV acts as a client. There are several limiting factors for WI-FI-based UAVs. For example, the operational range of a WI-FI-based UAV is typically limited to 150 feet (46 m) indoor and 300 feet (92 m) outdoor. There is significant latency for control and video behaviors. Interference by other WI-FI devices affects operational continuity of the WI-FI-based UAVs.

Demand in the UAV user community has made more professional-level protocols available in the commercial and retail markets. By way of example but not limitation, two common RC protocols used for UAVs are Lightbridge and OcuSync. Enhancements in drone technology inevitably increases the capability of drones for use in industrial espionage and as weapons for nefarious activities.

Lightbridge is developed for long range and reliable communication. Communication is available within a range up to 5 km. Lightbridge supports 8 selectable channels, and the selection can be manual or automatic. Drones with Lightbridge protocol also have the ability to assess interference and move to alternate channels for greater quality.

OcuSync is developed based on the LightBridge protocol. OcuSync uses effective digital compression and other improvements, which decreases knowledge required to operate. OcuSync provides reliable HD and UHD video, and OcuSync-based drones can be operated in areas with greater dynamic interference. Ocusync improves command and control efficiencies and reduces latency. With OcuSync, video communications are improved substantially, operational range is increased, command and control recovery are enhanced when interference occurs.

The systems and methods of the present invention for unmanned vehicle recognition are operable to detect and classify UAVs at a distance, provide directions of the UAVs, and take defensive measures to mitigate risks. The detection and classification are fast, which provides more time to react and respond to threats. Exact detection range is based upon selection of antenna systems, topology, morphology, and client criteria. Classification of the detected UAVs provides knowledge of the UAVs and defines effective actions and

capabilities for countering UAV threats. In one embodiment, the direction information of the UAVs provides orientation within the environment based on the location of the UAV detector.

In one embodiment, the systems and methods of the present invention provides unmanned vehicle recognition solution targeting radio controlled and WI-FI-based drones. The overall system is capable of surveying the spectrum from 20 MHz to 6 GHz, not just the common 2.4 GHz and 5.8 GHz areas as in the prior art. In one embodiment, the systems and methods of the present invention are applied to address 2 major categories: RC-based UAV systems and WI-FI-based UAV systems. In one embodiment, UAV systems utilize RC protocols comprising LightBridge and OcuSync. In another embodiment, UAV systems are WI-FI based, for example but not for limitation, 3DR Solo and Parrot SkyController. The systems and methods of the present invention are operable to detect UAVs and their controllers by protocol.

The systems and methods of the present invention maintain a state-of-the-art learning system and library for classifying detected signals by manufacturer and controller type. The state-of-the-art learning system and library are updated as new protocols emerge.

In one embodiment, classification by protocol chipset is utilized to provide valuable intelligence and knowledge for risk mitigation and threat defense. The valuable intelligence and knowledge include effective operational range, supported peripherals (e.g., external or internal camera, barometers, GPS and dead reckoning capabilities), integrated obstacle avoidance systems, and interference mitigation techniques.

The state-of-the-art learning system of the present invention is highly accurate and capable of assessing detected UAV signals and/or controller signals for classification in less than a few seconds with a high confidence level. The state-of-the-art learning system is operable to discriminate changes in the environment for non-drone signals as well as drone signals. FIG. 3 is an illustration of Phantom 4 controller and drone signals. A human interface is operable to display the classification results. FIG. 4 illustrates a human interface display for drone detection according to one embodiment of the present invention.

It is difficult to recognize commercial and retail drones with the naked eye over 100 meters. It is critical to obtain a vector to the target for situational awareness and defense execution. The systems and methods of the present invention provides lines of bearing for direction finding for multiple UAVs flying simultaneously. Each line of bearing is color coded for display. Angles, along with frequencies utilized for uplink and downlink, are also displayed on the human interface.

Once a UAV is detected and classified, an alert is posted to a counter UAV system operator (e.g., a network operation center, an individual operator) including azimuth of the UAV and other information. The alert is transmitted via email, short message service (SMS) or third-party system integration. The counter UAV system is operable to engage an intercession transmission, which will disrupt the communication between the UAV and its controller. When the communication between the UAV and its controller is intercepted, the UAV will invoke certain safety protocols, such as reduce height and hover, land, or return to the launch point. The counter UAV system may have certain restrictions based on country and classification of the UAV.

In one embodiment, the systems and methods of the present invention are operable to update the UAV library

with emerging protocols for classification purposes, and refine the learning engine for wideband spectrum analysis for other potential UAV signatures, emerging protocols and technologies. In other words, the systems and methods of the present invention are adaptable to any new and emerging protocols and technologies developed for unmanned vehicles. In one embodiment, multiple node devices in the present invention are deployed to operate as a group of networked nodes. In one embodiment, the group of networked nodes are operable to estimate geographical locations for unmanned vehicles. In one embodiment, two node devices are operable to provide a single line of bearing and approximate a geographical location of a detected drone or controller. The more node devices there are in the group of network nodes, the more lines of bearing are operable to be provided, and the more accurate the geographical location is estimated for the detected drone or controller. In one embodiment, the geolocation function provides altitude and distance of a targeted drone.

In one embodiment, the counter UAV system in the present invention is operable to alert when unexpected signal characteristics are detected in 2.4 GHz and 5.8 GHz areas and classify the unexpected signal characteristics as potential UAV activities. In another embodiment, the counter UAV system in the present invention is operable to alert when unexpected signal characteristics are detected anywhere from 20 MHz to 6 GHz and classify the unexpected signal characteristics as potential UAV activities. In another embodiment, the counter UAV system in the present invention is operable to classify the unexpected signal characteristics as potential UAV activities when unexpected signal characteristics are detected anywhere from 40 MHz to 6 GHz. The automatic signal detection engine and analytics engine are enhanced in the counter UAV system to recognize potential UAV activities across a great portion of the spectrum. In one embodiment, any blocks of spectrum from 40 MHz to 6 GHz are operable to be selected for UAV recognition.

In one embodiment, vector-based information including inclinations, declinations, topology deviations, and user configurable Northing map orientation is added to the WGS84 mapping system for direction finding and location estimation. In one embodiment, earth-centered earth-fixed vector analysis is provided for multi-node systems to estimate UAV locations, derive UAV velocities from position changes over time, and determine UAV trajectory vectors in fixed nodal processing. In one embodiment, a group of networked node devices are operable to continually provide lines of bearing over time, approximate geographical locations of a detected unmanned vehicle on or above the earth, and track the movement of the detected unmanned vehicle from one estimated location to another. In one embodiment, the group of networked node devices are operable to determine velocities of the detected unmanned vehicle based on estimated locations and travel time. In one embodiment, the group of networked node devices are operable to estimate a trajectory of the detected unmanned vehicle based on the estimated geographical locations over time. In one embodiment, the group of networked node devices are operable to estimate accelerations and decelerations of the unmanned vehicle based on the velocities of the unmanned vehicles over time.

In one embodiment, the systems and methods of the present invention are operable for UAV detection and direction finding for different modulation schemes including but not limited to DSSS, OFDM, FHSS, FASST, etc. In one embodiment, the counter UAV system in the present inven-

tion is configured with cameras for motion detection. The cameras have both day and night vision.

In one embodiment, systems and methods of the present invention provides training for unmanned vehicle recognition. RF data is captured for a Phantom 3 drone and its controller and a Phantom 4 drone and its controller, both of which use Lightbridge protocol. RF data is also captured for a Mavic Pro drone and its controller, which uses OcuSync protocol. The RF data is recorded at different channels, different RF bandwidths, and different video quality settings inside and outside an Anechoic Chamber. FIG. 5 shows a setup for RF data capture in an Anechoic Chamber according to one embodiment of the present invention. The recordings are overlaid on the RF environment, and fading and channel hopping are simulated. FIG. 6 illustrates a simulation for fading and channel hopping of an OcuSync drone.

In one embodiment, the recorded RF data is used to train and calibrate an inception based convolutional neural network comprised in a drone detection system. FIG. 7 is an illustration of an inception based convolutional neural network. U.S. Patent Publication No. 2018/0137406 titled "Efficient Convolutional Neural Networks and Techniques to Reduce Associated Computational Costs" is incorporated herein by reference in its entirety. The inception based convolutional neural network generates probabilities that drones or their controllers are detected. The detection probabilities are updated multiple times per second.

The trained inception based convolutional neural network is operable to identify Lightbridge 1 controller and drone, Lightbridge 2 controller and drone, and OcuSync controller and drone. The trained inception based convolutional neural network is operable to identify Lightbridge and Ocusync controllers and drones at the same time. In one embodiment, the drone detection system comprising the trained inception based convolutional neural network is operable to search an instantaneous bandwidth of 147.2 MHz.

In one embodiment, the drone detection system of the present invention includes an artificial intelligence (AI) algorithm running on a single board computer (e.g., Nvidia Jetson TX2) with an execution time less than 10 ms. The drone detection system is operable to separate Phantom 3 and Phantom 4 controllers. Waveforms for Phantom 3 and Phantom 4 controllers are sufficiently different to assign separate probabilities.

The Artificial Intelligence (AI) algorithm is used to enhance performance for RF data analytics. The RF data analytics process based on the AI algorithm is visualized. The RF waterfalls of several drone scenarios are presented in FIGS. 8-18. FIG. 8 illustrates a scenario for an RF environment with just noise. FIG. 9 illustrates a scenario for an RF environment with a Phantom 4 controller and drone. FIG. 10 illustrates a scenario for an RF environment with two Mavic Pro drones. FIG. 11 illustrates a scenario for an RF environment with a Mavic Pro controller only. FIG. 12 illustrates a scenario for an RF environment with one Mavic Pro drone only. FIG. 13 illustrates a scenario for an RF environment with one Phantom 3 controller only. FIG. 14 illustrates a scenario for an RF environment with a Phantom 3 controller and drone. FIG. 15 illustrates a scenario for an RF environment with a Mavic Pro controller and drone in wideband mode. FIG. 16 illustrates a scenario for an RF environment with a Mavic Pro drone in wideband mode. FIG. 17 illustrates a scenario for an RF environment with a Mavic Pro controller and drone. FIG. 18 illustrates a scenario for an RF environment with a Mavic Pro controller and a Phantom 4 controller.

Each scenario is illustrated with 6 waterfall images. Each waterfall represents ~80 ms of time and 125 MHz of bandwidth. The top left image is the waterfall before an AI processing. The other five images are waterfalls after the AI processing. For each signal type, the areas of the waterfall that are likely for the RF signal type are highlighted. Areas that are not for the signal type are grayed out. The overall probability that a signal exists in the image is printed in the title of each waterfall image. In one embodiment, the AI algorithm is securely integrated with a state engine and a detection process of the present invention. In one embodiment, AI processing or processing of the waterfall includes a comparison of the waterfall image to a database of waterfall images of the RF environment or similar RF environments for which signals or noise were identified. The comparison of the waterfall image to the database of waterfall images is operable to be performed in real time or near real time. In one embodiment, the database of waterfall images is operable to be updated in real time or near real time with the waterfall image created based on the at least one tile and associated information with the waterfall image, including but not limited to the signal identified in the waterfall image, the signal type(s) identified in the waterfall image, and/or noise identified in the waterfall image.

The comparison includes the use of machine learning (ML) or convolutional neural networks (CNN) in one embodiment. In other embodiments, the node or system is operable to utilize a plurality of learning techniques for analyzing waterfall images including, but not limited to, artificial intelligence (AI), deep learning (DL), artificial neural networks (ANNs), support vector machines (SVMs), Markov decision process (MDP), and/or natural language processing (NLP). The node or system is operable to use any of the aforementioned learning techniques alone or in combination. Further, the node or system is operable to utilize predictive analytics techniques including, but not limited to, machine learning (ML), artificial intelligence (AI), neural networks (NNs) (e.g., long short term memory (LSTM) neural networks), deep learning, historical data, and/or data mining to make future predictions and/or models. The node or system is preferably operable to recommend and/or perform actions based on historical data, external data sources, ML, AI, NNs, and/or other learning techniques. The node or system is operable to utilize predictive modeling and/or optimization algorithms including, but not limited to, heuristic algorithms, particle swarm optimization, genetic algorithms, technical analysis descriptors, combinatorial algorithms, quantum optimization algorithms, iterative methods, deep learning techniques, and/or feature selection techniques.

In one embodiment, a method for drone detection and classification comprises applying FFT function to RF data, converting FFT data into logarithmic scale in magnitude, averaging converted FFT into 256 by 256 array representing 125 MHz of bandwidth and 80 ms of time as a base tile, applying normalization function to the base tile, applying a series of convolutional and pooling layers, applying modified You Only Look Once (YOLO) algorithm for detection, grouping bounding boxes displayed in the waterfall images (e.g., waterfall plots in FIGS. 8-18), classifying signals based on the shape of detection output, verifying results with a second level recurrent neural network (RNN) based pattern estimator.

In one embodiment, a method for training comprises recording clean RF signals, shifting RF signals in frequency randomly, creating truth data for YOLO output, adding a simulated channel to the RF signals, recording typical RF

15

backgrounds, applying FFT function to RF data, converting FFT data into logarithmic scale in magnitude, averaging converted FFT into 256 by 256 array representing 125 MHz of bandwidth and 80 ms of time as a base tile, applying normalization function to the base tile, applying a series of convolutional and pooling layers, applying modified You Only Look Once (YOLO) algorithm for detection, grouping bounding boxes displayed in the waterfall images (e.g., waterfall plots in FIGS. 8-18), applying a sigmoid cross entropy function, and applying an Adaptive Moment Estimation (Adam) based back propagation algorithm.

In one embodiment, a drone detection engine is operable to convert FFT flows from a radio to a tile. For each channel, the drone detection engine is operable to standardize the FFT output from the radio at a defined resolution bandwidth, and group high resolution FFT data into distinct bins over time. The drone detection engine is further operable to calculate average and standard deviation of power for discrete FFT bins, assign a power value to each channel within the tile. Each scan or single stare at the radio is a time slice, and multiple time slices with power and channel assignment create a tile. Tiles from different frequency spans and center frequencies are identified as a tile group by a tile group number. Receivers in the drone detection system are operable to be re-tuned to different frequencies and spans. In one embodiment, the drone detection system comprises multiple receivers to generate tiles and tile groups.

In one embodiment, a tile is sent to a YOLO AI Engine. Outputs of a decision tree in the YOLO AI engine are used to detect multiple drones and their controllers. Drones of the same type of radio protocol are operable to be identified within the tile. Controllers of the same type of radio protocol are operable to be identified within the tile. Drones of different radio protocols are also operable to be identified within the tile. Controllers of different radio protocols are also operable to be identified within the tile. FIG. 19 is an illustration of identifying drones and controllers based on signal edge detection.

In one embodiment, a plurality of tiles is sent to the YOLO AI engine. In one embodiment, a tile group is sent to the YOLO AI engine. The YOLO AI engine generates an output for each tile to identify drones and their controllers with a probability. An average probability is calculated based on outputs for multiple tiles in the tile group. For each tile group, the YOLO AI engine computes outputs for several tiles per second.

In one embodiment, a state engine controls the flows of tiles and tile groups into one or more AI engines. AI engines do not use frequency values for analytics. Thus, the one or more AI engines are operable for any frequency and frequency span that a drone radio supports. The state engine further correlates output of the one or more AI engines to appropriate tiles and tile groups.

The systems and methods of the present invention are operable for direction finding of drones and their controllers. Outputs from the AI engine are denoted with time basis for the drones and their controllers.

Drones typically maintain the same frequency unless their firmware detects interference. Then the drones may negotiate a change with their controllers. This does not create an issue for detection as long as the new frequency and span is monitored by the systems and methods of the present invention. Drone controllers typically use a frequency hopping spread spectrum (FHSS) or other Frequency hopping system (e.g., Gaussian frequency shift keying (GFSK)).

In one embodiment, the systems and method of the present invention are operable to approximate a start time of

16

a line of bearing for a direction finding (DF) system. The time intervals are either known or estimated based upon the behavior monitored by the AI engine and state engine. This allows the time slice and frequency of each individual drone and/or controller to be passed to the DF system. In one embodiment, three or four receivers are coordinated to collect information in appropriate frequency segments, wherein the frequency segments are similar to tiles described earlier. FIG. 20 is an illustration with averaged signal amplitudes and signal edges according to one embodiment of the present invention.

The AI engine examines the segments to determine if a drone or controller exists. An azimuth of the drone or controller in an Earth-Centered Earth-Fixed coordinate system is determined based on other information collected from the three or four receivers using time difference of arrival (TDOA), angle of arrival (AOA), power correlative, or interferometry techniques.

Distance capability of UAV detection and classification system depends on hardware configuration, environment morphology and restrictions based on country and classification of the counter UAV operator. In one embodiment, the systems and methods for unmanned vehicle recognition are operable to detect unmanned vehicles within 3-4 kilometers. FIG. 21 displays a detection range of less than 500 meters based on equipment specification and location.

In one embodiment, the systems and methods of the present invention are operable to utilize training data. In one embodiment training data contains at least one tile displaying at least one waterfall image for one or more drones. These images are numbered based on the time slices and color coded to keep track of each drone. The numbering and color coding of each drone is fed into the ML drone frequency hopping algorithm. The color coding and numbering of each drone signal allows for easy training of the image analysis algorithm to quickly pick up on trends in drone frequency hopping which allows for quick training of the algorithm to account for multiple drones and/or frequency hopping types. In one embodiment, one or more drones contain the same and/or different types of radio protocols within at least one tile. In one embodiment the training data is further utilizable by one or more AI engines operable for any frequency and frequency span supported by drone radio. In one embodiment, the AI engine is operable to use machine learning (ML), artificial intelligence (AI), You Only Look Once Artificial Intelligence (YOLO AI), deep learning (DL), neural networks (NNs), artificial neural networks (ANNs), convolutional neural networks (CNNs), support vector machines (SVMs), Markov decision process (MDP), natural language processing (NLP), control theory, and/or statistical learning techniques. FIG. 22 displays a tile containing drones of the same type of radio protocol. FIG. 23 displays training data of a tile containing drones of different types of radio protocols.

In one embodiment the AI and/or ML algorithm is operable to classify the drone and/or signal modulation type based on the waterfall image created by the signal. The training data and/or waterfall images are operable to be sets of known modulation schemes for drones for the ML algorithm to provide an image comparison to the signal in the RF environment. The known frequency hopping schemes are operable to be learned by the algorithm and then identified by pattern recognition when a signal in the environment is analyzed and put into a waterfall graphical representation. The image comparison ML or AI algorithm is operable to detect the modulation type patterns of the signal.

17

In one embodiment, the training data is operable to be utilized for “No Drone Zone” identification and avoidance. The Federal Aviation Administration (FAA) denotes areas where drones are unable to operate as “No Drone Zones”, with each zone containing specific operating restrictions. To report “No Drone Zones” to recreational flyers, the FAA released B4UFLY—a service that allows recreational flyers to determine where drones are operable and not operable.

In one embodiment, the training data is operable to be fed to the ML Drone Frequency Hopping algorithm to allow for identification and avoidance of “No Drone Zones”. In a further embodiment, the ML Drone Frequency Hopping algorithm is operable to cross reference detected signals with B4UFLY to determine if the signal is in a restricted area.

In one embodiment, the training data includes at least a tile or group of tiles representing a drone that is operable to exist in a “No Drone Zone”. In one embodiment, the system utilizes the ML or AI Drone Frequency Hopping algorithm to confirm detection of a drone before cross referencing the signal with B4UFLY, the geolocation engine, and other sensors to determine if the drone has breached a “No Drone Zone”. In one embodiment, other sensors include visual systems such as LIDAR, cameras and radar. The system is operable to interface with the multiple data sources, determine if the information from the multiple data sources is consistent or in agreement, and create a report identifying and determining if the drone has breached a “No Drone Zone”. If the sensor information is not in agreement, the system is operable to discard the data and create a report identifying the faulty data.

In one embodiment, the present invention is operable to utilize training data to detect friendly drones (F-Drone) and nefarious drones (N-Drone). In one embodiment an F-Drone is identified by a specific radio protocol and/or a N-Drone is identified by not fitting into the desired radio protocol. In another embodiment, an N-Drone is identified by a desired radio protocol and/or a F-Drone is identified by not identifying with a targeted radio protocol. In a further embodiment one or more AI engines is operable to detect F-Drones and/or N-Drones utilizing training data. Training data is further utilizable by one or more AI engines that are operable for any frequency and frequency span supported by drone radio. In one embodiment, training data is operable to be utilized by one or more AI engines to define a “No Drone Zone”. In a further embodiment, the AI engine is operable to detect a N-Drones by identifying a signal within the “No Drone Zone”. In one embodiment training data is cross referenced with B4UFLY, the geolocation engine, and other sensors to embody parameters, including but not limited to altitude, to detect friendly and/or nefarious drones. In one embodiment, a report is created and sent identifying and detecting F-Drones and/or N-Drones.

Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

18

What is claimed is:

1. A system for signal identification in a radiofrequency (RF) environment, comprising:

at least one node device including a processor and memory in communication with at least one RF receiver;

wherein the at least one node device is in communication with at least one artificial intelligence (AI) engine;

wherein the at least one RF receiver is operable to capture RF data in the RF environment and transmit the RF data to the at least one node device;

wherein the at least one node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile;

wherein the at least one AI engine is operable to generate an output for the at least one tile to identify and correlate at least one signal with a probability;

wherein the at least one tile is represented as at least one waterfall image; and

wherein location information for the at least one signal is displayed on at least one user interface as at least one line of bearing.

2. The system of claim 1, wherein the at least one AI engine is operable to calculate an average probability based on the output.

3. The system of claim 1, wherein the at least one AI engine is trained using at least one training data set.

4. The system of claim 1, wherein a signal modulation type of the at least one signal includes direct sequence spread spectrum (DSSS), orthogonal frequency division multiplexing (OFDM), frequency hopping spread spectrum (FHSS), and/or Futaba advanced spread spectrum technology (FASST).

5. The system of claim 1, wherein the at least one signal includes a drone signal.

6. The system of claim 1, wherein the at least one node device is operable to transmit an alert and/or report related to a comparison of the location information of the at least one signal against at least one location of interest.

7. The system of claim 1, wherein the at least one line of bearing is calculated by a direction-finding engine.

8. The system of claim 1, wherein the at least one signal is highlighted on the at least one waterfall image.

9. The system of claim 1, wherein the at least one RF receiver is operable to transform the RF data to the FFT data.

10. An apparatus for signal identification in a radiofrequency (RF) environment, comprising:

a node device including a processor and memory;

wherein the node device is in communication with at least one artificial intelligence (AI) engine;

wherein the node device is operable to receive RF data from at least one RF receiver;

wherein the node device is operable to average Fast Fourier Transform (FFT) data derived from the RF data into at least one tile;

wherein the at least one AI engine is operable to generate an output for the at least one tile to identify and correspond at least one signal with a probability;

wherein the RF data includes location information of the at least one signal;

wherein the at least one tile is represented as at least one waterfall image; and

wherein the location information for the at least one signal is displayed on at least one user interface as at least one color coded line of bearing.

19

11. The apparatus of claim 10, wherein the location information includes an airport no-fly-zone and/or at least one geofence.

12. The apparatus of claim 10, wherein the at least one AI engine is operable to calculate an average probability based on the output. 5

13. The apparatus of claim 10, wherein the at least one node device is operable to transmit an alert and/or report related to a comparison of the location information of the at least one signal against at least one location of interest. 10

14. The apparatus of claim 10, wherein the at least one AI engine is operable to be trained using at least one training data set.

15. The apparatus of claim 10, wherein the at least one color coded line of bearing is calculated by a direction-finding engine. 15

16. A method for signal analysis in a radiofrequency (RF) environment, comprising:

capturing RF data in the RF environment by at least one RF receiver, converting the RF data to Fast Fourier Transform (FFT) data by the at least one RF receiver and transmitting the FFT data to at least one node device; 20

averaging the FFT data by the at least one node device into at least one tile and representing the at least one tile as at least one waterfall image; 25

generating an output for the at least one tile by at least one artificial intelligence (AI) engine in communication

20

with the at least one node device to identify and correlate at least one signal with a probability; calculating an average probability based on the output of the at least one tile;

classifying at least one signal emitting device based by comparing the at least one signal emitted from the at least one signal emitting device to classification data stored in a classification library; and

wherein location information for the at least one signal is displayed on at least one user interface as at least one line of bearing.

17. The method of claim 16, wherein the location information includes an airport no-fly-zone and/or at least one geofence.

18. The method of claim 16, further comprising the at least one node device transmitting an alert and/or report related to a comparison of the location information for the at least one signal against at least one location of interest.

19. The method of claim 16, wherein the at least one AI engine is trained using at least one training data set.

20. The method of claim 16, wherein the at least one signal characteristic includes a modulation type of the at least one signal that includes direct sequence spread spectrum (DSSS), orthogonal frequency division multiplexing (OFDM), frequency hopping spread spectrum (FHSS), and/or Futaba advanced spread spectrum technology (FASST).

* * * * *